

1. Questions

There are two series given below as series I and series II. Both series have one missing term as [X] and [Y] respectively. Both series follow different patterns.

Series I: 8, 18, 57, __[X]__, 1165, 6996

Series II: 12, 16, 32, 68, __[Y]__, 232

Which of the following is/are definitely true?

- I). The difference between [X] and [Y] is a perfect square.
- II). Both [X] and [Y] are exactly divisible by 4.
- III). The sum of [X] and [Y] is more than 200 but less than 400.
- a. Only I
- b. Both I and II
- c. Both II and III
- d. All I, II and III
- e. None is true

2. Questions

There are two series given below as series I and series II. Series I has one wrong term, that will be replaced by the correct term, which is equal to [P]. Both series follow the same pattern.

Series I: 5, 12, 33, 70, 117, 180

Series II: __[P/17]__, ____, ____, ____, ____, __[3Q+26]__

Determine the correct relation between [P] and [Q].

- a. $P > Q$
- b. $P < Q$
- c. $P \geq Q$
- d. $P \leq Q$
- e. $P = Q$

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3. Questions

There are three series given below as series I, series II and series III. Each series has one missing term as [A], [B] and [C] respectively. Each series follows the same pattern.

Series I: 12, 15, 9, 18, __[A]__, 21

Series II: __[A+4]__, 13, 7, 16, 4, __[B]__

Series III: __[B-8]__, 14, __[C]__, 17, 5, 20

Find the value of $[(A+C)-B]$.

- a. 5
- b. -5
- c. 10
- d. -10
- e. 15

4. Questions

There is a series given below, which has one missing term as [M] and one wrong term as [N]

Series: 16, 6, __[M]__, 5, 11, 4, 10

Which of the following is/are definitely true?

I). The sum of the digits of [M] is equal to [N]

II). [N] is the lowest two-digit prime number.

III). [M] is an exact multiple of 7.

- a. Only III
- b. Both I and II
- c. Both II and III
- d. All I, II and III
- e. None is true

5. Questions

There is a series given below, which has three missing terms as [R], [S] and [T]. (S,T are perfect squares)

Series: 28, 30, 27, [R], [S], [T]

Which of the following equation is correct?

- a. $[2R-S] = 39$
- b. $[T-R] = 5$
- c. $[S+3T] = 130$
- d. All of the above
- e. None of these

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6. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below shows the ratio of the quantity of milk and water in the initial mixture of four different vessels A, B, C and D. It also shows the percentage of the initial mixture is taken out, the total quantity of mixture (milk + water) added (in ml) in the remaining mixture and the ratio of the quantity of milk and water in the resultant mixture in each vessel.

Vessels	The ratio of the quantity of milk and water in the initial mixture	The percentage of the initial mixture is taken out	The total quantity of mixture (milk + water) added in the remaining mixture (in ml)	The ratio of the quantity of milk and water in the resultant mixture
A	2:1	25%	15	13:8
B	3:2	40%	10	8:5
C	2:1	20%	11	21:10
D	5:3	50%	30	3:2

Note:

The total quantity of mixture = the quantity of milk + the quantity of water

The ratio of the quantity of milk and water in the remaining mixture will remain the same as the initial mixture.

The ratio of the quantity of water added in the remaining mixture of vessel A and B is 5:1.

The quantity of milk added in the remaining mixture of vessel B is 3 ml more than that of vessel A.

The quantity of milk added in the remaining mixture of vessel C is 60% of vessel D.

The difference between the quantity of water added in the remaining mixture of vessel C and D is 13 ml. (C < D)

If _____ % of the initial mixture of vessel A is taken out and replaced with the same quantity of water, then the ratio of the quantity of milk and water in the resultant mixture of vessel A will be _____.

Which of the following does satisfy the filler(s) in the same order?

I). 40 and 2:3

II). 15 and 3:5

III). 30 and 7:8

- Only I follow
- Both II and III follow
- Both I and III follow
- All I, II and III follow
- None follow

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7. Questions

If the resultant mixture of vessel D is further poured into another vessel, in which the ratio of the initial quantity of milk and water is 3:1, then the quantity of milk will be 80 ml more than water in the final mixture. Find the total initial mixture (milk + water) in another mixture.

- a. 100 ml
- b. 144 ml
- c. 128 ml
- d. 160 ml
- e. None of these

8. Questions

The total initial mixture of another vessel E is twice of vessel C and the ratio of the quantity of milk and water in the initial mixture of vessel E is $[X:Y]$. If 11.11% of the initial mixture of vessel E is taken out and replaced with 16 ml of water, then the quantity of milk will be 48 ml more than water in the resultant mixture of vessel E.

Which of the following equation(s) is/ are prime number?

I). $[X-Y]$

II). $[3Y/X]$

III). $[X \times Y]$

- a. Both II and III follow
- b. Only II follow
- c. Both I and III follow
- d. All I, II and III follow
- e. None follow

9. Questions

If the initial mixture of vessel B and C are mixed together, then $[Z]\%$ of milk is replaced with $[4Z+4]$ ml of alcohol, so that the total quantity of the resultant mixture (milk + water + alcohol) becomes 400 ml.

Which of the following represents the value of $[Z]$?

- a. Exactly divisible by 20
- b. Even number
- c. Lowest composite number
- d. Both a) and b)

e. None of these

10. Questions

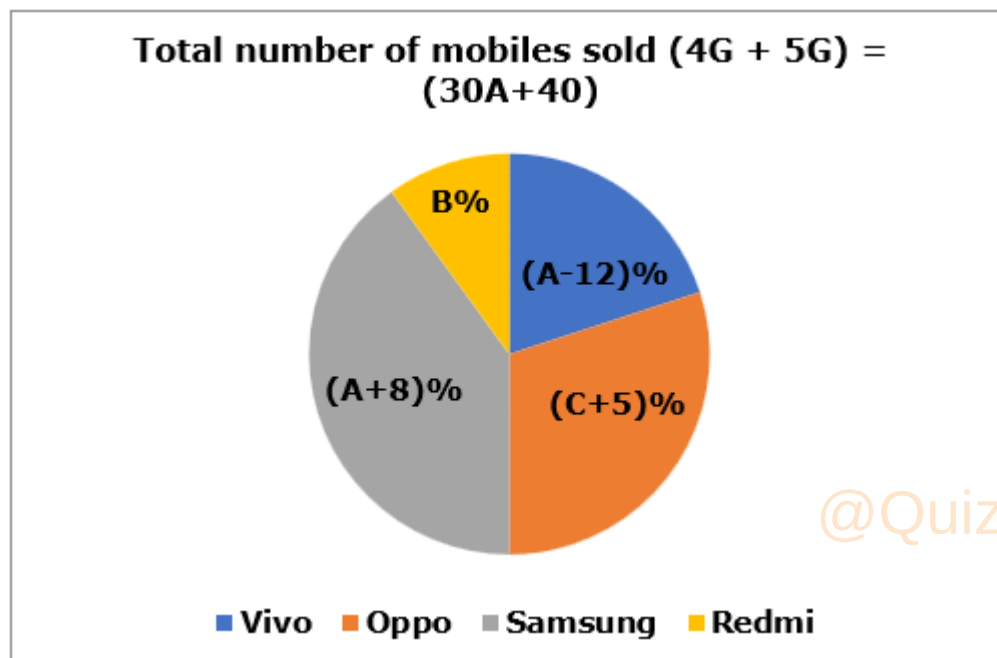
Another vessel F contains a mixture of 150 ml of milk and water in the ratio of the 4:1. Now, 20% of the initial mixture is taken out and added milk and water in the ratio of 2:3, so that the quantity of water in the resultant mixture becomes 30% of milk. Find the total quantity of milk and water added in the remaining mixture of vessel F.

- a. 10 ml
- b. 12 ml
- c. 20 ml
- d. 24 ml
- e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of mobiles sold (4G + 5G) by four companies namely Vivo, Oppo, Samsung and Redmi. The table chart given below shows the total number of mobiles manufactured (4G + 5G), the ratio of the number of 4G mobiles sold and 5G mobiles unsold and the difference between the number of 4G and 5G mobiles unsold ($4G - 5G$) by each company.



Company	The total number of mobiles manufactured (4G + 5G)	The ratio of the number of 4G mobiles sold and 5G mobiles unsold	The difference between the number of 4G and 5G mobiles unsold (4G – 5G)
Vivo	250	6:1	10
Oppo	460	8:3	40
Samsung	475	12:1	25
Redmi	155	15:2	35

Note:

All the variables given in the pie chart are positive integers.

The total number of mobiles manufactured = the number of 4G mobiles manufactured + the number of 5G mobiles manufactured

The total number of 4G mobiles manufactured = the number of 4G mobiles sold + the number of 4G mobiles unsold

The total number of 5G mobiles manufactured = the number of 5G mobiles sold + the number of 5G mobiles unsold

The number of 4G mobiles unsold by each company is more than the number of 5G mobiles unsold by the same company.

The total number of mobiles sold (4G + 5G) by all four companies together is (30A+40).

The total number of mobiles sold (4G + 5G) by Vivo is 200, which is equal to the difference between the total number of mobiles sold (4G + 5G) by Oppo and Redmi. (Oppo > Redmi)+

Out of the total number of 5G mobiles manufactured by Vivo, [X]% are defective and rest are non-defective. The number of non-defective mobiles which remained unsold is equal to the number of defective mobiles which are sold. (Only non-defective mobiles can be sold)

Which of the following is/are definitely true about [X] ?

I). The smallest composite number, which is exactly divisible by 2.

II). A two-digit natural number.

III). The unit digit is a whole number.

- Only II
- Both II and III
- Both I and II
- All I, II and III
- None is true

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12. Questions

The number of 5G mobiles sold to females by Oppo is $\frac{P}{Q}$, which is 80% of the number of 4G mobiles sold to females by the same company. If the ratio of the number of 4G mobiles sold to males and females by Oppo is 9:7, then the number of 5G mobiles sold to males by Oppo will be $\frac{P}{Q}$.

- a. $P > Q$
- b. $P < Q$
- c. $P \geq Q$
- d. $P \leq Q$
- e. $P = Q$

13. Questions

The cost price of each 4G mobile of Redmi is Rs. $\frac{700D-400}{Q}$ and the overall selling price of all 4G mobiles sold by Redmi is Rs. 900000. If the discount allowed on each 4G mobile is Rs. 2000 and each 4G mobile is marked-up by 75% above its cost price, then find the overall cost price of all 4G mobiles unsold by Redmi.

- a. Rs. 400000
- b. Rs. 280000
- c. Rs. 360000
- d. Cannot be determined
- e. None of these

14. Questions

If the total number of mobiles sold (4G + 5G) by another company Apple is 60% of Samsung, out of which $\frac{3}{4}$ are 4G, while the ratio of the number of 4G and 5G mobiles which remained unsold by Apple is 1:2 and the total number of 5G mobiles manufactured by Apple is 50% of 4G mobiles manufactured by the same company, then find the total number of mobiles manufactured (4G + 5G) by Apple.

- a. 300
- b. 280
- c. 440
- d. 500
- e. None of these

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15. Questions

If the number of 3G mobiles unsold by Samsung is $\frac{Y}{Q}$ % of the 4G mobiles unsold by the same

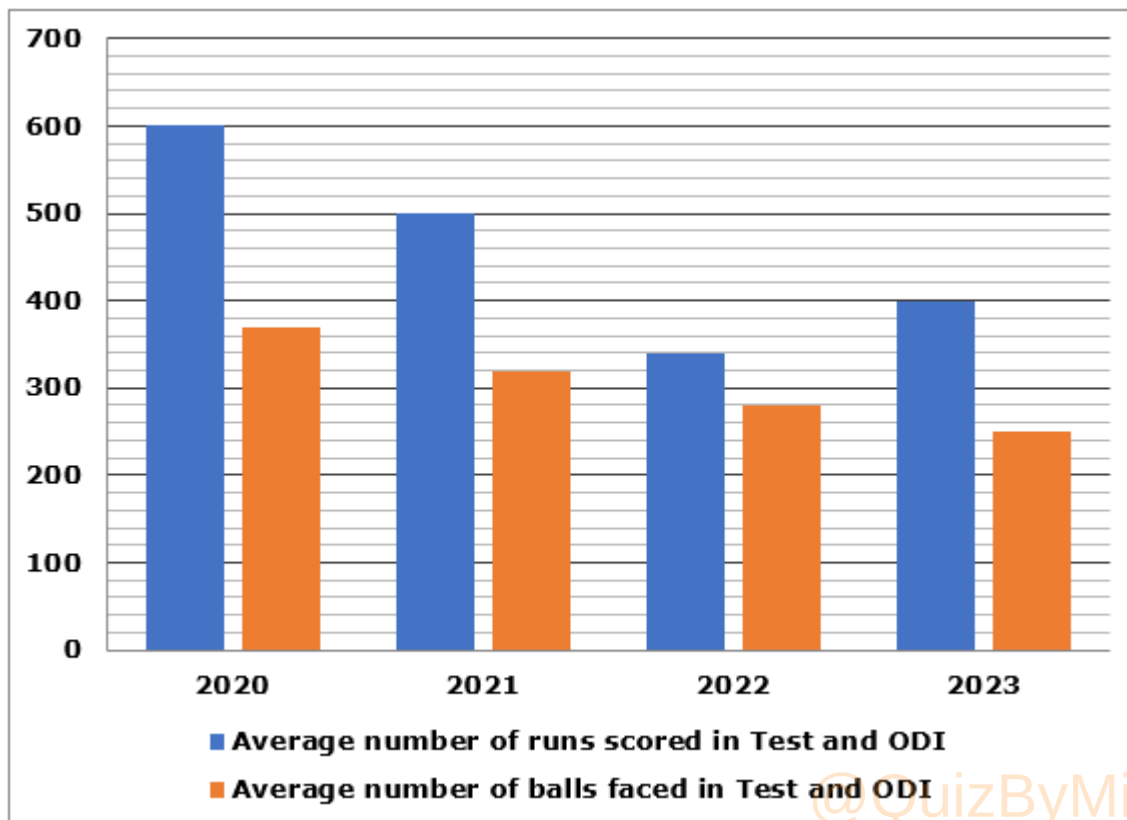
company and the number of 3G mobiles sold by Samsung is 80, then the total number of mobiles manufactured (3G + 4G+ 5G) by Samsung will be increased by $[2Y+20]$. Find the value of $[Y/2]$.

- a. 40
- b. 35
- c. 55
- d. 20
- e. None of these

16. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the average number of runs scored by a player in Tests and ODIs in four different years 2020, 2021, 2022 and 2023. It also shows the average number of balls faced by the player in Test and ODI in each year.



Note:

The total number of runs scored = the number of runs scored in Tests + the number of runs scored in ODIs

The total number of balls faced = the number of balls faced in Tests + the number of balls faced in ODIs

The strike-rate = $\left[\left\{ \frac{\text{the total number of runs scored}}{\text{the total number of balls faced}} \right\} \times 100 \right]$

The number of runs scored in Test in 2021 is $\frac{4}{5}$ th of the previous year.

The number of balls faced in Test in 2020 is 640, which is 60% more than 2021.

The strike-rate in Test in 2020 is 35 less than 2021.

The ratio of the number of runs scored in ODIs in 2022 and 2023 is 9:16.

The number of balls faced in ODIs in 2023 is 100, which is 40 more than the previous year.

The sum of the strike-rate in ODIs in 2022 and 2023 together is 620.

In 2020, out of the total number of balls faced by the player in ODIs, the number of dot balls faced by him is [4A]. If the number of dot balls faced by him is excluded, then his strike-rate will be [100A]. Find the number of dot balls faced by the player is what percent of the total number of balls faced by him in ODIs in 2020? (A is single-digit natural number)

- a. 36%
- b. 12%
- c. 45%
- d. 20%
- e. None of these

17. Questions

In 2023, the total number of innings played by the player in Test is [X]. If the total number of runs made by the player in the next 4 innings is 200, then his average in Test will be increased by 4. Find the total number of matches played by the player in Test in 2023. (Each Test match has 2 innings and X is a perfect square)

- a. 16
- b. 18
- c. 8
- d. 9
- e. None of these

18. Questions

In 2021, if the number of sixes scored by the player in ODIs is [P], which is $\frac{1}{5}$ th of the number of fours scored by the player in ODI in the same year and the number of runs scored other than boundaries in ODIs in 2021 is [16P] less than the total number of runs scored in boundaries (fours + sixes), then the value of P will be ____[?]____ number.

What should be come in place of question mark?

- a. Composite
- b. Prime
- c. Odd
- d. All of the above

e. None of these

19. Questions

In 2022, if the number of runs scored by another player in Test is $[Z]\%$ more than the given player, while the number of balls faced by another is same as the number of balls faced by the given player, then the strike-rate of another player in Test in 2022 will be $[3Z+20]$. Find the number of runs scored by another player in Test in 2022.

- a. 700
- b. 640
- c. 480
- d. Cannot be determined
- e. None of these

20. Questions

In another year 2024, the total number of runs scored (Test + ODIs) by the player is 25% more than the previous year and the number of balls faced by the player in ODIs is 125, which is 75% less than Test. If the strike-rate of the player in Test in 2024 is 50 less than ODIs in the same year, then find the ratio of the number of runs scored by the player in Test and ODIs in 2024.

- a. 1:3
- b. 3:1
- c. 3:2
- d. 2:3
- e. None of these

Explanations:

1. Questions

Answer: D

Series I: 8, 18, 57, $[X]$, 1165, 6996

1st term = 8

2nd term = $(8 \times 2 + 2) = (16 + 2) = 18$

3rd term = $(18 \times 3 + 3) = (54 + 3) = 57$

4th term = $(57 \times 4 + 4) = (228 + 4) = \mathbf{232 (X)}$

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$$5^{\text{th}} \text{ term} = (232 \times 5 + 5) = (1160 + 5) = 1165$$

$$6^{\text{th}} \text{ term} = (1165 \times 6 + 6) = (6990 + 6) = 6996$$

$$\text{So, } X = 232$$

Series II: 12, 16, 32, 68, [Y], 232

$$1^{\text{st}} \text{ term} = 12$$

$$2^{\text{nd}} \text{ term} = (12 + 2^2) = (12 + 4) = 16$$

$$3^{\text{rd}} \text{ term} = (16 + 4^2) = (16 + 16) = 32$$

$$4^{\text{th}} \text{ term} = (32 + 6^2) = (32 + 36) = 68$$

$$5^{\text{th}} \text{ term} = (68 + 8^2) = (68 + 64) = \mathbf{132 (Y)}$$

$$6^{\text{th}} \text{ term} = (132 + 10^2) = (132 + 100) = 232$$

$$\text{So, } Y = 132$$

$$\text{I. } (X - Y) = (232 - 132) = 100$$

100 is a perfect square of 10.

So, I is definitely true.

$$\text{II. } (X/4) = (232/4) = 58$$

$$(Y/4) = (132/4) = 33$$

Both X and Y are exactly divisible by 4.

So, II is definitely true.

$$\text{III. } (X + Y) = (232 + 132) = 364$$

364 is more than 200 but less than 400.

So, III is definitely true.

Therefore, all I, II and III are definitely true.

Hence, the correct answer is option D.

2. Questions

Answer: A

Series I: 5, 12, 33, 70, 117, 180

$$1^{\text{st}} \text{ term} = 5$$

$$2^{\text{nd}} \text{ term} = (5 + 7) = 12$$

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$$3^{\text{rd}} \text{ term} = (12+21) = 33$$

$$4^{\text{th}} \text{ term} = (33+35) = 68 \text{ (70 is wrong)}$$

$$5^{\text{th}} \text{ term} = (68+49) = 117$$

$$6^{\text{th}} \text{ term} = (117+63) = 180$$

So, the wrong term = 70

And, the correct term = P = 68

Series II: $\frac{P}{17}$, ____, ____, ____, ____, $3Q+26$

$\frac{68}{17}$, ____, ____, ____, ____, $3Q+26$

4, ____, ____, ____, ____, $3Q+26$

$$1^{\text{st}} \text{ term} = 4$$

$$2^{\text{nd}} \text{ term} = (4+7) = 11$$

$$3^{\text{rd}} \text{ term} = (11+21) = 32$$

$$4^{\text{th}} \text{ term} = (32+35) = 67$$

$$5^{\text{th}} \text{ term} = (67+49) = 116$$

$$6^{\text{th}} \text{ term} = (116+63) = 179$$

Now,

$$(3Q+26) = 179$$

$$3Q = 153$$

$$Q = 51$$

Now, P = 68

And Q = 51

Therefore, $P > Q$.

Hence, the correct answer is option A.

3. Questions

Answer: B

Series I: 12, 15, 9, 18, A , 21

$$1^{\text{st}} \text{ term} = 12$$

$$2^{\text{nd}} \text{ term} = (12+3) = 15$$

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$$3^{\text{rd}} \text{ term} = (15-6) = 9$$

$$4^{\text{th}} \text{ term} = (9+9) = 18$$

$$5^{\text{th}} \text{ term} = (18-12) = \mathbf{6 \text{ (A)}}$$

$$6^{\text{th}} \text{ term} = (6+15) = 21$$

So, the missing term = A = 6

Series II: $_\text{[A+4]}_\$, 13, 7, 16, 4, $_\text{[B]}_\$

$_\text{[6+4]}_\$, 13, 7, 16, 4, $_\text{[B]}_\$

10, 13, 7, 16, 4, $_\text{[B]}_\$

$$1^{\text{st}} \text{ term} = 10$$

$$2^{\text{nd}} \text{ term} = (10+3) = 13$$

$$3^{\text{rd}} \text{ term} = (13-6) = 7$$

$$4^{\text{th}} \text{ term} = (7+9) = 16$$

$$5^{\text{th}} \text{ term} = (16-12) = 4$$

$$6^{\text{th}} \text{ term} = (4+15) = \mathbf{19 \text{ (B)}}$$

So, the missing term = B = 19

Series III: $_\text{[B-8]}_\$, 14, $_\text{[C]}_\$, 17, 5, 20

$_\text{[19-8]}_\$, 14, $_\text{[C]}_\$, 17, 5, 20

11, 14, $_\text{[C]}_\$, 17, 5, 20

$$1^{\text{st}} \text{ term} = 11$$

$$2^{\text{nd}} \text{ term} = (11+3) = 14$$

$$3^{\text{rd}} \text{ term} = (14-6) = 8 \text{ (C)}$$

$$4^{\text{th}} \text{ term} = (8+9) = 17$$

$$5^{\text{th}} \text{ term} = (17-12) = 5$$

$$6^{\text{th}} \text{ term} = (5+15) = 20$$

So, the missing term = C = 8

Therefore, the value of $\{(A+C)-B\} = \{(6+8)-19\} = (14-19) = -5$.

Hence, the correct answer is option B.

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4. Questions

Answer: C

Series: 16, 6, __[M]__, 5, 11, 4

1st term = 16

2nd term = $(16/2-2) = (8-2) = 6$

3rd term = $(6 \times 2 + 2) = (12 + 2) = 14$ (**M**)

4th term = $(14/2-2) = (7-2) = 5$

5th term = $(5 \times 2 + 2) = (10 + 2) = 12$ (**11 is wrong**)

6th term = $(12/2-2) = (6-2) = 4$

7th term = $(4 \times 2 + 2) = 8 + 2 = 10$

So, the missing term = M = 14

And, the wrong term = N = 11

I). The sum of the digits of M = $(1+4) = 5$

But, N = $(1+1) = 2$

So, I is definitely not true.

II). 11 is the lowest two-digit prime number.

So, II is definitely true.

III). 14 is exactly divisible by 7.

So, III is definitely true.

Therefore, both II and III are definitely true.

Hence, the correct answer is option C.

5. Questions

Answer: A

Series: 28, 30, 27, [R], [S], [T]

1st term = 28

2nd term = $(28+2) = 30$

3rd term = $(30-3) = 27$

4th term = $(27+5) = 32$ (**R**)

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$$5^{\text{th}} \text{ term} = (32-7) = \mathbf{25 \text{ (S)}}$$

$$6^{\text{th}} \text{ term} = (25+11) = \mathbf{36 \text{ (T)}}$$

So,

$$R = 32$$

$$S = 25$$

$$T = 36$$

$$(2R-S) = (2 \times 32 - 25) = (64 - 25) = 39$$

So, option a) does follow.

$$(T-R) = (36-32) = 4$$

So, option b) does not follow.

$$(S+3T) = (25+3 \times 36) = (25+108) = 133$$

So, option c) does not follow.

Therefore, only option a) is correct.

Hence, the correct answer is option A.

6. Questions

The total quantity of the mixture (milk + water) added in vessel A = 15 ml

The total quantity of the mixture (milk + water) added in vessel B = 10 ml

Let the quantity of water added in the remaining mixture of vessel A and B be 5m ml and m ml respectively.

So, the quantity of milk added in the remaining mixture of vessel A = (15-5m) ml

And, the quantity of milk added in the remaining mixture of vessel B = (10-m) ml

Now,

$$(10-m) - (15-5m) = 3$$

$$7 - m = 15 - 5m$$

$$4m = 8$$

$$m = 2$$

Now,

The total quantity of the mixture (milk + water) added in vessel C = 11 ml

The total quantity of the mixture (milk + water) added in vessel D = 30 ml

The ratio of the quantity of milk added in the remaining mixture of vessel C and D = 60:100 = 3:5. Let it be 3n ml and 5n ml.

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So, the quantity of water added in the remaining mixture of vessel C = $(11-3n)$ ml

And, the quantity of water added in the remaining mixture of vessel D = $(30-5n)$ ml

Now,

$$(30-5n) - (11-3n) = 13$$

$$17 - 5n = 11 - 3n$$

$$2n = 6$$

$$n = 3$$

Vessel A

Let the initial mixture be $300P$ ml

So, the initial quantity of milk = $\frac{2}{3}$ of $300P = 200P$ ml

And, the initial quantity of water = $\frac{1}{3}$ of $300P = 100P$ ml

Now,

The total quantity of remaining mixture = 75% of $300P$ ml = $225P$ ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of $225P = 150P$ ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of $225P = 75P$ ml

The quantity of milk added in the remaining mixture = $(15-5m) = (15-5 \times 2) = (15-10) = 5$ ml

The quantity of water added in the remaining mixture = $5m = 10$ ml

Now,

$$(150P+5)/(75P+10) = 13/8$$

$$1200P + 40 = 975P + 130$$

$$225P = 90$$

$$P = 0.40$$

Therefore,

The initial mixture (milk + water) = $300P = (300 \times 0.40) = 120$ ml

The initial quantity of milk = $200P = (200 \times 0.40) = 80$ ml

The initial quantity of water = $100P = (100 \times 0.40) = 40$ ml

Vessel B

Let the initial mixture be $500Q$ ml

So, the initial quantity of milk = $\frac{3}{5}$ of $500Q = 300Q$ ml

And, the initial quantity of water = $\frac{2}{5}$ of $500Q = 200Q$ ml

Now,

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The total quantity of remaining mixture = 60% of 500Q ml = 300Q ml

The quantity of milk in the remaining mixture = $\frac{3}{5}$ of 300Q = 180Q ml

The quantity of water in the remaining mixture = $\frac{2}{5}$ of 300Q = 120Q ml

The quantity of milk added in the remaining mixture = $(10-m) = (10-2) = 8$ ml

The quantity of water added in the remaining mixture = $m = 2$ ml

Now,

$$(180Q+8)/(120Q+2) = 8/5$$

$$900Q + 40 = 960Q + 16$$

$$60Q = 24$$

$$Q = 0.40$$

Therefore,

The initial mixture (milk + water) = 500Q = $(500 \times 0.40) = 200$ ml

The initial quantity of milk = 300Q = $(300 \times 0.40) = 120$ ml

The initial quantity of water = 200Q = $(200 \times 0.40) = 80$ ml

Vessel C

Let the initial mixture be 300R ml

So, the initial quantity of milk = $\frac{2}{3}$ of 300R = 200R ml

And, the initial quantity of water = $\frac{1}{3}$ of 300R = 100R ml

Now,

The total quantity of remaining mixture = 80Q of 300R ml = 240R ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of 240R = 160R ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of 240R = 80R ml

The quantity of milk added in the remaining mixture = $3n = (3 \times 3) = 9$ ml

The quantity of water added in the remaining mixture = $(11-3n) = (11-9) = 2$ ml

Now,

$$(160R+9)/(80R+2) = 21/10$$

$$1600R + 90 = 1680R + 42$$

$$80R = 48$$

$$R = 0.60$$

Therefore,

The initial mixture (milk + water) = 300R = $(300 \times 0.60) = 180$ ml

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The initial quantity of milk = $200R = (200 \times 0.60) = 120$ ml

The initial quantity of water = $100R = (100 \times 0.60) = 60$ ml

Vessel D

Let the initial mixture be $800S$ ml

So, the initial quantity of milk = $\frac{5}{8}$ of $800S = 500S$ ml

And, the initial quantity of water = $\frac{3}{8}$ of $800S = 300S$ ml

Now,

The total quantity of remaining mixture = 50% of $800S = 400S$ ml

The quantity of milk in the remaining mixture = $\frac{5}{8}$ of $400S = 250S$ ml

The quantity of water in the remaining mixture = $\frac{3}{8}$ of $400S = 150S$ ml

The quantity of milk added in the remaining mixture = $5n = (5 \times 3) = 15$ ml

The quantity of water added in the remaining mixture = $(30 - 5n) = (30 - 15) = 15$ ml

Now,

$$(250S + 15) / (150S + 15) = 3/2$$

$$500S + 30 = 450S + 45$$

$$50S = 15$$

$$S = 0.30$$

Therefore,

The initial mixture (milk + water) = $800S = (800 \times 0.30) = 240$ ml

The initial quantity of milk = $500S = (500 \times 0.30) = 150$ ml

The initial quantity of water = $300S = (300 \times 0.30) = 90$ ml

Vessels	The total initial mixture (in ml)	The quantity of milk in initial mixture (in ml)	The quantity of water in initial mixture (in ml)
A	120	80	40
B	200	120	80
C	180	120	60
D	240	150	90
Total	740	470	270

Answer: C

Now, according to the question,

Vessel A

The initial mixture = 120 ml

The initial quantity of milk = 80 ml

The initial quantity of water = 40 ml

Using I

The quantity of initial mixture taken out = 40% of 120 = 48 ml

So, the resultant quantity of mixture = $(120 - 48) = 72$ ml

The resultant quantity of milk = $\frac{2}{3}$ of 72 = 48 ml

The resultant quantity of water = $\{(1/3 \text{ of } 72) + 48\} = (24 + 48) = 72$ ml

The ratio of the quantity of milk and water in the resultant mixture = $48:72 = 2:3$.

Therefore, I does satisfy.

Using II

The quantity of initial mixture taken out = 15% of 120 = 18 ml

So, the resultant quantity of mixture = $(120 - 18) = 102$ ml

The resultant quantity of milk = $\frac{2}{3}$ of 102 = 68 ml

The resultant quantity of water = $\{(1/3 \text{ of } 102) + 18\} = (34 + 18) = 52$ ml

The ratio of the quantity of milk and water in the resultant mixture = $68:52 = 17:13$.

Therefore, II does not satisfy.

Using III

The quantity of initial mixture taken out = 30% of 120 = 36 ml

So, the resultant quantity of mixture = $(120 - 36) = 84$ ml

The resultant quantity of milk = $\frac{2}{3}$ of 84 = 56 ml

The resultant quantity of water = $\{(1/3 \text{ of } 84) + 36\} = (28 + 36) = 64$ ml

The ratio of the quantity of milk and water in the resultant mixture = $56:64 = 7:8$.

Therefore, III does satisfy.

So, both I and III follow.

Hence, the correct answer is option C.

7. Questions

The total quantity of the mixture (milk + water) added in vessel A = 15 ml

The total quantity of the mixture (milk + water) added in vessel B = 10 ml

Let the quantity of water added in the remaining mixture of vessel A and B be $5m$ ml and m ml respectively.

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So, the quantity of milk added in the remaining mixture of vessel A = $(15-5m)$ ml

And, the quantity of milk added in the remaining mixture of vessel B = $(10-m)$ ml

Now,

$$(10-m) - (15-5m) = 3$$

$$7 - m = 15 - 5m$$

$$4m = 8$$

$$m = 2$$

Now,

The total quantity of the mixture (milk + water) added in vessel C = 11 ml

The total quantity of the mixture (milk + water) added in vessel D = 30 ml

The ratio of the quantity of milk added in the remaining mixture of vessel C and D = $60:100 = 3:5$. Let it be $3n$ ml and $5n$ ml.

So, the quantity of water added in the remaining mixture of vessel C = $(11-3n)$ ml

And, the quantity of water added in the remaining mixture of vessel D = $(30-5n)$ ml

Now,

$$(30-5n) - (11-3n) = 13$$

$$17 - 5n = 11 - 3n$$

$$2n = 6$$

$$n = 3$$

Vessel A

Let the initial mixture be $300P$ ml

So, the initial quantity of milk = $\frac{2}{3}$ of $300P = 200P$ ml

And, the initial quantity of water = $\frac{1}{3}$ of $300P = 100P$ ml

Now,

The total quantity of remaining mixture = 75% of $300P$ ml = $225P$ ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of $225P = 150P$ ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of $225P = 75P$ ml

The quantity of milk added in the remaining mixture = $(15-5m) = (15-5 \times 2) = (15-10) = 5$ ml

The quantity of water added in the remaining mixture = $5m = 10$ ml

Now,

$$(150P+5)/(75P+10) = 13/8$$

$$1200P + 40 = 975P + 130$$

$$225P = 90$$

$$P = 0.40$$

Therefore,

$$\text{The initial mixture (milk + water)} = 300P = (300 \times 0.40) = 120 \text{ ml}$$

$$\text{The initial quantity of milk} = 200P = (200 \times 0.40) = 80 \text{ ml}$$

$$\text{The initial quantity of water} = 100P = (100 \times 0.40) = 40 \text{ ml}$$

Vessel B

Let the initial mixture be 500Q ml

$$\text{So, the initial quantity of milk} = \frac{3}{5} \text{ of } 500Q = 300Q \text{ ml}$$

$$\text{And, the initial quantity of water} = \frac{2}{5} \text{ of } 500Q = 200Q \text{ ml}$$

Now,

$$\text{The total quantity of remaining mixture} = 60\% \text{ of } 500Q \text{ ml} = 300Q \text{ ml}$$

$$\text{The quantity of milk in the remaining mixture} = \frac{3}{5} \text{ of } 300Q = 180Q \text{ ml}$$

$$\text{The quantity of water in the remaining mixture} = \frac{2}{5} \text{ of } 300Q = 120Q \text{ ml}$$

$$\text{The quantity of milk added in the remaining mixture} = (10 - m) = (10 - 2) = 8 \text{ ml}$$

$$\text{The quantity of water added in the remaining mixture} = m = 2 \text{ ml}$$

Now,

$$(180Q + 8) / (120Q + 2) = 8/5$$

$$900Q + 40 = 960Q + 16$$

$$60Q = 24$$

$$Q = 0.40$$

Therefore,

$$\text{The initial mixture (milk + water)} = 500Q = (500 \times 0.40) = 200 \text{ ml}$$

$$\text{The initial quantity of milk} = 300Q = (300 \times 0.40) = 120 \text{ ml}$$

$$\text{The initial quantity of water} = 200Q = (200 \times 0.40) = 80 \text{ ml}$$

Vessel C

Let the initial mixture be 300R ml

$$\text{So, the initial quantity of milk} = \frac{2}{3} \text{ of } 300R = 200R \text{ ml}$$

$$\text{And, the initial quantity of water} = \frac{1}{3} \text{ of } 300R = 100R \text{ ml}$$

Now,

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The total quantity of remaining mixture = 80Q of 300R ml = 240R ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of 240R = 160R ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of 240R = 80R ml

The quantity of milk added in the remaining mixture = $3n = (3 \times 3) = 9$ ml

The quantity of water added in the remaining mixture = $(11-3n) = (11-9) = 2$ ml

Now,

$$(160R+9)/(80R+2) = 21/10$$

$$1600R + 90 = 1680R + 42$$

$$80R = 48$$

$$R = 0.60$$

Therefore,

The initial mixture (milk + water) = $300R = (300 \times 0.60) = 180$ ml

The initial quantity of milk = $200R = (200 \times 0.60) = 120$ ml

The initial quantity of water = $100R = (100 \times 0.60) = 60$ ml

Vessel D

Let the initial mixture be 800S ml

So, the initial quantity of milk = $\frac{5}{8}$ of 800S = 500S ml

And, the initial quantity of water = $\frac{3}{8}$ of 800S = 300S ml

Now,

The total quantity of remaining mixture = 50% of 800S = 400S ml

The quantity of milk in the remaining mixture = $\frac{5}{8}$ of 400S = 250S ml

The quantity of water in the remaining mixture = $\frac{3}{8}$ of 400S = 150S ml

The quantity of milk added in the remaining mixture = $5n = (5 \times 3) = 15$ ml

The quantity of water added in the remaining mixture = $(30-5n) = (30-15) = 15$ ml

Now,

$$(250S+15)/(150S+15) = 3/2$$

$$500S + 30 = 450S + 45$$

$$50S = 15$$

$$S = 0.30$$

Therefore,

The initial mixture (milk + water) = $800S = (800 \times 0.30) = 240$ ml

@QuizByMickey

The initial quantity of milk = 500S = $(500 \times 0.30) = 150$ ml

The initial quantity of water = 300S = $(300 \times 0.30) = 90$ ml

Vessels	The total initial mixture (in ml)	The quantity of milk in initial mixture (in ml)	The quantity of water in initial mixture (in ml)
A	120	80	40
B	200	120	80
C	180	120	60
D	240	150	90
Total	740	470	270

Answer: A

Vessel D

The quantity of milk in the resultant mixture = $(250S+15) = (250 \times 0.30+15) = (75+15) = 90$ ml

The quantity of water in the resultant mixture = $(150S+15) = (150 \times 0.30+15) = (45+15) = 60$ ml

Let the initial quantity of milk and water in another mixture be 3k ml and k ml respectively.

Now,

$$(90+3k) - (60+k) = 80$$

$$10 + 3k = 60 + k$$

$$2k = 50$$

$$k = 25$$

Therefore, the total initial mixture (milk + water) in another vessel = $(3k+k) = 4k = (4 \times 25) = 100$ ml.

Hence, the correct answer is option A.

8. Questions

The total quantity of the mixture (milk + water) added in vessel A = 15 ml

The total quantity of the mixture (milk + water) added in vessel B = 10 ml

Let the quantity of water added in the remaining mixture of vessel A and B be 5m ml and m ml respectively.

So, the quantity of milk added in the remaining mixture of vessel A = $(15-5m)$ ml

And, the quantity of milk added in the remaining mixture of vessel B = $(10-m)$ ml

Now,

$$(10-m) - (15-5m) = 3$$

$$7 - m = 15 - 5m$$

$$4m = 8$$

$$m = 2$$

Now,

The total quantity of the mixture (milk + water) added in vessel C = 11 ml

The total quantity of the mixture (milk + water) added in vessel D = 30 ml

The ratio of the quantity of milk added in the remaining mixture of vessel C and D = 60:100 = 3:5. Let it be 3n ml and 5n ml.

So, the quantity of water added in the remaining mixture of vessel C = (11-3n) ml

And, the quantity of water added in the remaining mixture of vessel D = (30-5n) ml

Now,

$$(30-5n) - (11-3n) = 13$$

$$17 - 5n = 11 - 3n$$

$$2n = 6$$

$$n = 3$$

Vessel A

Let the initial mixture be 300P ml

So, the initial quantity of milk = $\frac{2}{3}$ of 300P = 200P ml

And, the initial quantity of water = $\frac{1}{3}$ of 300P = 100P ml

Now,

The total quantity of remaining mixture = 75% of 300P ml = 225P ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of 225P = 150P ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of 225P = 75P ml

The quantity of milk added in the remaining mixture = (15-5m) = (15-5×2) = (15-10) = 5 ml

The quantity of water added in the remaining mixture = 5m = 10 ml

Now,

$$(150P+5)/(75P+10) = 13/8$$

$$1200P + 40 = 975P + 130$$

$$225P = 90$$

$$P = 0.40$$

Therefore,

The initial mixture (milk + water) = 300P = (300×0.40) = 120 ml

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The initial quantity of milk = $200P = (200 \times 0.40) = 80$ ml

The initial quantity of water = $100P = (100 \times 0.40) = 40$ ml

Vessel B

Let the initial mixture be $500Q$ ml

So, the initial quantity of milk = $\frac{3}{5}$ of $500Q = 300Q$ ml

And, the initial quantity of water = $\frac{2}{5}$ of $500Q = 200Q$ ml

Now,

The total quantity of remaining mixture = 60% of $500Q$ ml = $300Q$ ml

The quantity of milk in the remaining mixture = $\frac{3}{5}$ of $300Q = 180Q$ ml

The quantity of water in the remaining mixture = $\frac{2}{5}$ of $300Q = 120Q$ ml

The quantity of milk added in the remaining mixture = $(10-m) = (10-2) = 8$ ml

The quantity of water added in the remaining mixture = $m = 2$ ml

Now,

$$(180Q+8)/(120Q+2) = 8/5$$

$$900Q + 40 = 960Q + 16$$

$$60Q = 24$$

$$Q = 0.40$$

Therefore,

The initial mixture (milk + water) = $500Q = (500 \times 0.40) = 200$ ml

The initial quantity of milk = $300Q = (300 \times 0.40) = 120$ ml

The initial quantity of water = $200Q = (200 \times 0.40) = 80$ ml

Vessel C

Let the initial mixture be $300R$ ml

So, the initial quantity of milk = $\frac{2}{3}$ of $300R = 200R$ ml

And, the initial quantity of water = $\frac{1}{3}$ of $300R = 100R$ ml

Now,

The total quantity of remaining mixture = 80% of $300R$ ml = $240R$ ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of $240R = 160R$ ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of $240R = 80R$ ml

The quantity of milk added in the remaining mixture = $3n = (3 \times 3) = 9$ ml

The quantity of water added in the remaining mixture = $(11-3n) = (11-9) = 2$ ml

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Now,

$$(160R+9)/(80R+2) = 21/10$$

$$1600R + 90 = 1680R + 42$$

$$80R = 48$$

$$R = 0.60$$

Therefore,

$$\text{The initial mixture (milk + water)} = 300R = (300 \times 0.60) = 180 \text{ ml}$$

$$\text{The initial quantity of milk} = 200R = (200 \times 0.60) = 120 \text{ ml}$$

$$\text{The initial quantity of water} = 100R = (100 \times 0.60) = 60 \text{ ml}$$

Vessel D

Let the initial mixture be $800S$ ml

$$\text{So, the initial quantity of milk} = 5/8 \text{ of } 800S = 500S \text{ ml}$$

$$\text{And, the initial quantity of water} = 3/8 \text{ of } 800S = 300S \text{ ml}$$

Now,

$$\text{The total quantity of remaining mixture} = 50\% \text{ of } 800S = 400S \text{ ml}$$

$$\text{The quantity of milk in the remaining mixture} = 5/8 \text{ of } 400S = 250S \text{ ml}$$

$$\text{The quantity of water in the remaining mixture} = 3/8 \text{ of } 400S = 150S \text{ ml}$$

$$\text{The quantity of milk added in the remaining mixture} = 5n = (5 \times 3) = 15 \text{ ml}$$

$$\text{The quantity of water added in the remaining mixture} = (30-5n) = (30-15) = 15 \text{ ml}$$

Now,

$$(250S+15)/(150S+15) = 3/2$$

$$500S + 30 = 450S + 45$$

$$50S = 15$$

$$S = 0.30$$

Therefore,

$$\text{The initial mixture (milk + water)} = 800S = (800 \times 0.30) = 240 \text{ ml}$$

$$\text{The initial quantity of milk} = 500S = (500 \times 0.30) = 150 \text{ ml}$$

$$\text{The initial quantity of water} = 300S = (300 \times 0.30) = 90 \text{ ml}$$

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Vessels	The total initial mixture (in ml)	The quantity of milk in initial mixture (in ml)	The quantity of water in initial mixture (in ml)
A	120	80	40
B	200	120	80
C	180	120	60
D	240	150	90
Total	740	470	270

Answer: B

The total initial mixture of vessel C = 180 ml

Vessel E

The total initial mixture = $(180 \times 2) = 360$ ml

The quantity of initial mixture is taken out = $1/9$ of $360 = 40$ ml (**Note: $11.11\% = 1/9$**)

The total quantity of resultant mixture after adding 16 ml of water = $\{(360-40)+16\} = (320+16) = 336$ ml

So, the quantity of milk in the resultant mixture = $\{(336+48)/2\} = (384/2) = 192$ ml

And, the quantity of water in the resultant mixture = $\{(336-48)/2\} = (288/2) = 144$ ml

Therefore, the quantity of water in the remaining mixture before adding 16 ml water = $(144-16) = 128$ ml

And, the quantity of milk in the remaining mixture before adding 16 ml water = 192 ml

The ratio of the quantity of milk and water in the remaining mixture = $192:128 = 3:2$.

The ratio of the quantity of milk and water in the remaining mixture will remain same as initial mixture. (**from note**)

Therefore, the ratio the quantity of milk and water in the initial mixture = $3:2$.

$$3:2 = X:Y$$

$$X = 3$$

$$Y = 2$$

$$\text{I. } (X-Y) = (3-2) = 1$$

1 is not a prime number.

So, I does not satisfy.

$$\text{II. } (3Y/X) = \{(3 \times 2)/3\} = (6/3) = 2$$

2 is a prime number.

So, II does satisfy.

$$\text{III. } (X \times Y) = (3 \times 2) = 6$$

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6 is not a prime number.

So, III does not satisfy.

Therefore, only II follow.

Hence, the correct answer is option B.

9. Questions

The total quantity of the mixture (milk + water) added in vessel A = 15 ml

The total quantity of the mixture (milk + water) added in vessel B = 10 ml

Let the quantity of water added in the remaining mixture of vessel A and B be $5m$ ml and m ml respectively.

So, the quantity of milk added in the remaining mixture of vessel A = $(15-5m)$ ml

And, the quantity of milk added in the remaining mixture of vessel B = $(10-m)$ ml

Now,

$$(10-m) - (15-5m) = 3$$

$$7 - m = 15 - 5m$$

$$4m = 8$$

$$m = 2$$

Now,

The total quantity of the mixture (milk + water) added in vessel C = 11 ml

The total quantity of the mixture (milk + water) added in vessel D = 30 ml

The ratio of the quantity of milk added in the remaining mixture of vessel C and D = $60:100 = 3:5$. Let it be $3n$ ml and $5n$ ml.

So, the quantity of water added in the remaining mixture of vessel C = $(11-3n)$ ml

And, the quantity of water added in the remaining mixture of vessel D = $(30-5n)$ ml

Now,

$$(30-5n) - (11-3n) = 13$$

$$17 - 5n = 11 - 3n$$

$$2n = 6$$

$$n = 3$$

Vessel A

Let the initial mixture be $300P$ ml

So, the initial quantity of milk = $\frac{2}{3}$ of $300P = 200P$ ml

And, the initial quantity of water = $\frac{1}{3}$ of $300P = 100P$ ml

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Now,

The total quantity of remaining mixture = 75% of 300P ml = 225P ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of 225P = 150P ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of 225P = 75P ml

The quantity of milk added in the remaining mixture = $(15-5m) = (15-5 \times 2) = (15-10) = 5$ ml

The quantity of water added in the remaining mixture = 5m = 10 ml

Now,

$$(150P+5)/(75P+10) = 13/8$$

$$1200P + 40 = 975P + 130$$

$$225P = 90$$

$$P = 0.40$$

Therefore,

The initial mixture (milk + water) = 300P = $(300 \times 0.40) = 120$ ml

The initial quantity of milk = 200P = $(200 \times 0.40) = 80$ ml

The initial quantity of water = 100P = $(100 \times 0.40) = 40$ ml

Vessel B

Let the initial mixture be 500Q ml

So, the initial quantity of milk = $\frac{3}{5}$ of 500Q = 300Q ml

And, the initial quantity of water = $\frac{2}{5}$ of 500Q = 200Q ml

Now,

The total quantity of remaining mixture = 60% of 500Q ml = 300Q ml

The quantity of milk in the remaining mixture = $\frac{3}{5}$ of 300Q = 180Q ml

The quantity of water in the remaining mixture = $\frac{2}{5}$ of 300Q = 120Q ml

The quantity of milk added in the remaining mixture = $(10-m) = (10-2) = 8$ ml

The quantity of water added in the remaining mixture = m = 2 ml

Now,

$$(180Q+8)/(120Q+2) = 8/5$$

$$900Q + 40 = 960Q + 16$$

$$60Q = 24$$

$$Q = 0.40$$

Therefore,

The initial mixture (milk + water) = $500Q = (500 \times 0.40) = 200$ ml

The initial quantity of milk = $300Q = (300 \times 0.40) = 120$ ml

The initial quantity of water = $200Q = (200 \times 0.40) = 80$ ml

Vessel C

Let the initial mixture be $300R$ ml

So, the initial quantity of milk = $\frac{2}{3}$ of $300R = 200R$ ml

And, the initial quantity of water = $\frac{1}{3}$ of $300R = 100R$ ml

Now,

The total quantity of remaining mixture = $80Q$ of $300R$ ml = $240R$ ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of $240R = 160R$ ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of $240R = 80R$ ml

The quantity of milk added in the remaining mixture = $3n = (3 \times 3) = 9$ ml

The quantity of water added in the remaining mixture = $(11 - 3n) = (11 - 9) = 2$ ml

Now,

$$(160R + 9) / (80R + 2) = 21/10$$

$$1600R + 90 = 1680R + 42$$

$$80R = 48$$

$$R = 0.60$$

Therefore,

The initial mixture (milk + water) = $300R = (300 \times 0.60) = 180$ ml

The initial quantity of milk = $200R = (200 \times 0.60) = 120$ ml

The initial quantity of water = $100R = (100 \times 0.60) = 60$ ml

Vessel D

Let the initial mixture be $800S$ ml

So, the initial quantity of milk = $\frac{5}{8}$ of $800S = 500S$ ml

And, the initial quantity of water = $\frac{3}{8}$ of $800S = 300S$ ml

Now,

The total quantity of remaining mixture = 50% of $800S = 400S$ ml

The quantity of milk in the remaining mixture = $\frac{5}{8}$ of $400S = 250S$ ml

The quantity of water in the remaining mixture = $\frac{3}{8}$ of $400S = 150S$ ml

The quantity of milk added in the remaining mixture = $5n = (5 \times 3) = 15$ ml

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The quantity of water added in the remaining mixture = $(30-5n) = (30-15) = 15$ ml

Now,

$$(250S+15)/(150S+15) = 3/2$$

$$500S + 30 = 450S + 45$$

$$50S = 15$$

$$S = 0.30$$

Therefore,

$$\text{The initial mixture (milk + water)} = 800S = (800 \times 0.30) = 240 \text{ ml}$$

$$\text{The initial quantity of milk} = 500S = (500 \times 0.30) = 150 \text{ ml}$$

$$\text{The initial quantity of water} = 300S = (300 \times 0.30) = 90 \text{ ml}$$

Vessels	The total initial mixture (in ml)	The quantity of milk in initial mixture (in ml)	The quantity of water in initial mixture (in ml)
A	120	80	40
B	200	120	80
C	180	120	60
D	240	150	90
Total	740	470	270

Answer: B

$$\text{The total initial quantity of milk in vessels B and C} = (120+120) = 240 \text{ ml}$$

$$\text{The total initial quantity of water in vessels B and C} = (80+60) = 140 \text{ ml}$$

$$\text{Now, the quantity of milk in the resultant mixture} = \{240 - (Z\% \text{ of } 240)\} = (240 - 2.40Z)$$

$$\text{And, the quantity of water in the resultant mixture} = 140 \text{ ml}$$

$$\text{The quantity of alcohol in the resultant mixture} = (4Z+4)$$

Now,

$$(240 - 2.40Z) + 140 + (4Z+4) = 400$$

$$1.60Z = 16$$

$$Z = 10$$

So, 10 is an even number.

Hence, the correct answer is option B.

10. Questions

$$\text{The total quantity of the mixture (milk + water) added in vessel A} = 15 \text{ ml}$$

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The total quantity of the mixture (milk + water) added in vessel B = 10 ml

Let the quantity of water added in the remaining mixture of vessel A and B be 5m ml and m ml respectively.

So, the quantity of milk added in the remaining mixture of vessel A = (15-5m) ml

And, the quantity of milk added in the remaining mixture of vessel B = (10-m) ml

Now,

$$(10-m) - (15-5m) = 3$$

$$7 - m = 15 - 5m$$

$$4m = 8$$

$$m = 2$$

Now,

The total quantity of the mixture (milk + water) added in vessel C = 11 ml

The total quantity of the mixture (milk + water) added in vessel D = 30 ml

The ratio of the quantity of milk added in the remaining mixture of vessel C and D = 60:100 = 3:5. Let it be 3n ml and 5n ml.

So, the quantity of water added in the remaining mixture of vessel C = (11-3n) ml

And, the quantity of water added in the remaining mixture of vessel D = (30-5n) ml

Now,

$$(30-5n) - (11-3n) = 13$$

$$17 - 5n = 11 - 3n$$

$$2n = 6$$

$$n = 3$$

Vessel A

Let the initial mixture be 300P ml

So, the initial quantity of milk = $\frac{2}{3}$ of 300P = 200P ml

And, the initial quantity of water = $\frac{1}{3}$ of 300P = 100P ml

Now,

The total quantity of remaining mixture = 75% of 300P ml = 225P ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of 225P = 150P ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of 225P = 75P ml

The quantity of milk added in the remaining mixture = (15-5m) = (15-5×2) = (15-10) = 5 ml

The quantity of water added in the remaining mixture = 5m = 10 ml

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Now,

$$(150P+5)/(75P+10) = 13/8$$

$$1200P + 40 = 975P + 130$$

$$225P = 90$$

$$P = 0.40$$

Therefore,

$$\text{The initial mixture (milk + water)} = 300P = (300 \times 0.40) = 120 \text{ ml}$$

$$\text{The initial quantity of milk} = 200P = (200 \times 0.40) = 80 \text{ ml}$$

$$\text{The initial quantity of water} = 100P = (100 \times 0.40) = 40 \text{ ml}$$

Vessel B

Let the initial mixture be 500Q ml

$$\text{So, the initial quantity of milk} = 3/5 \text{ of } 500Q = 300Q \text{ ml}$$

$$\text{And, the initial quantity of water} = 2/5 \text{ of } 500Q = 200Q \text{ ml}$$

Now,

$$\text{The total quantity of remaining mixture} = 60\% \text{ of } 500Q \text{ ml} = 300Q \text{ ml}$$

$$\text{The quantity of milk in the remaining mixture} = 3/5 \text{ of } 300Q = 180Q \text{ ml}$$

$$\text{The quantity of water in the remaining mixture} = 2/5 \text{ of } 300Q = 120Q \text{ ml}$$

$$\text{The quantity of milk added in the remaining mixture} = (10-m) = (10-2) = 8 \text{ ml}$$

$$\text{The quantity of water added in the remaining mixture} = m = 2 \text{ ml}$$

Now,

$$(180Q+8)/(120Q+2) = 8/5$$

$$900Q + 40 = 960Q + 16$$

$$60Q = 24$$

$$Q = 0.40$$

Therefore,

$$\text{The initial mixture (milk + water)} = 500Q = (500 \times 0.40) = 200 \text{ ml}$$

$$\text{The initial quantity of milk} = 300Q = (300 \times 0.40) = 120 \text{ ml}$$

$$\text{The initial quantity of water} = 200Q = (200 \times 0.40) = 80 \text{ ml}$$

Vessel C

Let the initial mixture be 300R ml

$$\text{So, the initial quantity of milk} = 2/3 \text{ of } 300R = 200R \text{ ml}$$

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And, the initial quantity of water = $\frac{1}{3}$ of $300R = 100R$ ml

Now,

The total quantity of remaining mixture = $\frac{80}{100}$ of $300R$ ml = $240R$ ml

The quantity of milk in the remaining mixture = $\frac{2}{3}$ of $240R = 160R$ ml

The quantity of water in the remaining mixture = $\frac{1}{3}$ of $240R = 80R$ ml

The quantity of milk added in the remaining mixture = $3n = (3 \times 3) = 9$ ml

The quantity of water added in the remaining mixture = $(11-3n) = (11-9) = 2$ ml

Now,

$$(160R+9)/(80R+2) = 21/10$$

$$1600R + 90 = 1680R + 42$$

$$80R = 48$$

$$R = 0.60$$

Therefore,

The initial mixture (milk + water) = $300R = (300 \times 0.60) = 180$ ml

The initial quantity of milk = $200R = (200 \times 0.60) = 120$ ml

The initial quantity of water = $100R = (100 \times 0.60) = 60$ ml

Vessel D

Let the initial mixture be $800S$ ml

So, the initial quantity of milk = $\frac{5}{8}$ of $800S = 500S$ ml

And, the initial quantity of water = $\frac{3}{8}$ of $800S = 300S$ ml

Now,

The total quantity of remaining mixture = 50% of $800S = 400S$ ml

The quantity of milk in the remaining mixture = $\frac{5}{8}$ of $400S = 250S$ ml

The quantity of water in the remaining mixture = $\frac{3}{8}$ of $400S = 150S$ ml

The quantity of milk added in the remaining mixture = $5n = (5 \times 3) = 15$ ml

The quantity of water added in the remaining mixture = $(30-5n) = (30-15) = 15$ ml

Now,

$$(250S+15)/(150S+15) = 3/2$$

$$500S + 30 = 450S + 45$$

$$50S = 15$$

$$S = 0.30$$

Therefore,

The initial mixture (milk + water) = $800S = (800 \times 0.30) = 240$ ml

The initial quantity of milk = $500S = (500 \times 0.30) = 150$ ml

The initial quantity of water = $300S = (300 \times 0.30) = 90$ ml

Vessels	The total initial mixture (in ml)	The quantity of milk in initial mixture (in ml)	The quantity of water in initial mixture (in ml)
A	120	80	40
B	200	120	80
C	180	120	60
D	240	150	90
Total	740	470	270

Answer: A

Vessel F

The total initial quantity of mixture = 150 ml

The initial quantity of milk = $\frac{4}{5}$ of 150 = 120 ml

The initial quantity of water = $\frac{1}{5}$ of 150 = 30 ml

Remaining mixture (vessel F)

The total quantity of mixture = 80% of 150 = 120 ml

The quantity of milk = $\frac{4}{5}$ of 120 = 96 ml

The quantity of water = $\frac{1}{5}$ of 120 = 24 ml

Now, let the quantity of milk and water added in the remaining mixture be 2T ml and 3T ml respectively.

So,

$$(24+3T)/(96+2T) = 30/100$$

$$(24+3T)/(96+2T) = 3/10$$

$$240 + 30T = 288 + 6T$$

$$24T = 48$$

$$T = 2$$

Therefore, the total quantity of mixture added in the mixture = $(2T+3T) = 5T = (5 \times 2) = 10$ ml.

Hence, the correct answer is option A.

11. Questions

From the pie chart given above, we have,

@QuizByMickey

The total value of the pie chart = 100%

The total number of mobiles sold by Vivo = $(A-12)\%$ of $(30A+40) = \{(30A^2+40A-360A-480)/100\} = \{(30A^2-320A-480)/100\}$

Now,

$$\{(30A^2-320A-480)/100\} = 200$$

$$30A^2 - 320A - 480 = 20000$$

$$30A^2 - 320A - 20480 = 0$$

$$3A^2 - 32A - 2048 = 0$$

$$3A^2 - 96A + 64A - 2048 = 0$$

$$3A(A-32) + 64(A-32) = 0$$

$$(3A+64)(A-32) = 0$$

$$A = -21.33 \text{ or } +32$$

So, $A = 32$ (positive integer)

Therefore, the total number of mobiles sold $(4G + 5G)$ by all four companies together = $(30A+40) = (30 \times 32 + 40) = (960 + 40) = 1000$

The total number of mobiles sold by Vivo = 200

The total number of mobiles sold by Samsung = $(A+8)\%$ of 1000 = $(32+8)\%$ of 1000 = 40% of 1000 = 400

Now, the percentage of the total number of mobiles sold by Vivo and Samsung together = $\{(A-12)+(A+8)\}\% = \{(32-12)+(32+8)\}\% = (20+40)\% = 60\%$

So, the percentage of the total number of mobiles sold by Oppo and Redmi together = $(100-60)\% = 40\%$

The total number of mobiles sold by Oppo = $(C+5)\%$ of 1000 = $(10C+50)$

And, the total number of mobiles sold by Redmi = $B\%$ of 1000 = $10B$

Now,

$$(10C+50) + 10B = 40\% \text{ of } 1000$$

$$(10C+50) + 10B = 400$$

$$10C + 10B = 350$$

$$C + B = 35 \dots \dots \dots (I)$$

Also,

$$(10C+50) - 10B = 200$$

$$10C - 10B = 150$$

@QuizByMickey

$$C - B = 15 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$2B = 20$$

$$B = 10$$

Now, put the value of B in equation (I), we get,

$$C + B = 35$$

$$C + 10 = 35$$

$$C = 25$$

Therefore,

$$\text{The total number of mobiles sold by Oppo} = (10C+50) = (10 \times 25 + 50) = (250+50) = 300$$

$$\text{And, the total number of mobiles sold by Redmi} = 10B = (10 \times 10) = 100$$

Vivo

$$\text{The total number of mobiles manufactured} = 250$$

$$\text{The total number of mobiles sold} = 200$$

$$\text{The total number of mobiles unsold} = (250-200) = 50$$

$$\text{The number of 4G mobiles unsold} = \{(50+10)/2\} = (60/2) = 30$$

$$\text{The number of 5G mobiles unsold} = \{(50-10)/2\} = (40/2) = 20$$

$$\text{So, the number of 4G mobiles sold} = \{20/1\} \times 6 = 120$$

$$\text{Therefore, the number of 5G mobiles sold} = (200-120) = 80$$

$$\text{Now, the total number of 4G mobiles manufactured} = (120+30) = 150$$

$$\text{And, the total number of 5G mobiles manufactured} = (80+20) = 100$$

Oppo

$$\text{The total number of mobiles manufactured} = 460$$

$$\text{The total number of mobiles sold} = 300$$

$$\text{The total number of mobiles unsold} = (460-300) = 160$$

$$\text{The number of 4G mobiles unsold} = \{(160+40)/2\} = (200/2) = 100$$

$$\text{The number of 5G mobiles unsold} = \{(160-40)/2\} = (120/2) = 60$$

$$\text{So, the number of 4G mobiles sold} = \{60/3\} \times 8 = 160$$

$$\text{Therefore, the number of 5G mobiles sold} = (300-160) = 140$$

$$\text{Now, the total number of 4G mobiles manufactured} = (160+100) = 260$$

$$\text{And, the total number of 5G mobiles manufactured} = (140+60) = 200$$

@QuizByMickey

Samsung

The total number of mobiles manufactured = 475

The total number of mobiles sold = 400

The total number of mobiles unsold = $(475-400) = 75$

The number of 4G mobiles unsold = $\{(75+25)/2\} = (100/2) = 50$

The number of 5G mobiles unsold = $\{(75-25)/2\} = (50/2) = 25$

So, the number of 4G mobiles sold = $\{25/1\} \times 12 = 300$

Therefore, the number of 5G mobiles sold = $(400-300) = 100$

Now, the total number of 4G mobiles manufactured = $(300+50) = 350$

And, the total number of 5G mobiles manufactured = $(100+25) = 125$

Redmi

The total number of mobiles manufactured = 155

The total number of mobiles sold = 100

The total number of mobiles unsold = $(155-100) = 55$

The number of 4G mobiles unsold = $\{(55+35)/2\} = (90/2) = 45$

The number of 5G mobiles unsold = $\{(55-35)/2\} = (20/2) = 10$

So, the number of 4G mobiles sold = $\{10/2\} \times 15 = 75$

Therefore, the number of 5G mobiles sold = $(100-75) = 25$

Now, the total number of 4G mobiles manufactured = $(75+45) = 120$

And, the total number of 5G mobiles manufactured = $(25+10) = 35$

4G Mobiles

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	150	120	30
Oppo	260	160	100
Samsung	350	300	50
Redmi	120	75	45

5G Mobiles

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	100	80	20
Oppo	200	140	60
Samsung	125	100	25
Redmi	35	25	10

Answer: B

Now, according to the question,

Vivo (5G mobiles)

The total number of mobiles manufactured = 100

The number of mobiles sold = 80 (all non-defective)

The number of mobiles unsold = $(100-80) = 20$

The number of defective mobiles = $X\%$ of 100 = X

The number of non-defective mobiles = $(100-X)$

The number of non-defective mobiles sold = 80

So, the number of non-defective mobiles unsold = $\{(100-X)-80\} = (20-X)$

Now,

$$(20-X) = 0$$

$$X = 20$$

I). 4, 6 and 8 are composite numbers, which are exactly divisible by 2 and they all are smaller than 20.

So, I is definitely not true.

II). 20 is a two-digit natural number.

So, II is definitely true.

III). The unit digit of 20 is 0.

0 is a whole number.

So, III is definitely true.

Therefore, both II and III are definitely true.

Hence, the correct answer is option B.

12. Questions

From the pie chart given above, we have,

@QuizByMickey

The total value of the pie chart = 100%

The total number of mobiles sold by Vivo = $(A-12)\%$ of $(30A+40) = \{(30A^2+40A-360A-480)/100\} = \{(30A^2-320A-480)/100\}$

Now,

$$\{(30A^2-320A-480)/100\} = 200$$

$$30A^2 - 320A - 480 = 20000$$

$$30A^2 - 320A - 20480 = 0$$

$$3A^2 - 32A - 2048 = 0$$

$$3A^2 - 96A + 64A - 2048 = 0$$

$$3A(A-32) + 64(A-32) = 0$$

$$(3A+64)(A-32) = 0$$

$$A = -21.33 \text{ or } +32$$

So, $A = 32$ (positive integer)

Therefore, the total number of mobiles sold $(4G + 5G)$ by all four companies together = $(30A+40) = (30 \times 32 + 40) = (960 + 40) = 1000$

The total number of mobiles sold by Vivo = 200

The total number of mobiles sold by Samsung = $(A+8)\%$ of 1000 = $(32+8)\%$ of 1000 = 40% of 1000 = 400

Now, the percentage of the total number of mobiles sold by Vivo and Samsung together = $\{(A-12)+(A+8)\}\% = \{(32-12)+(32+8)\}\% = (20+40)\% = 60\%$

So, the percentage of the total number of mobiles sold by Oppo and Redmi together = $(100-60)\% = 40\%$

The total number of mobiles sold by Oppo = $(C+5)\%$ of 1000 = $(10C+50)$

And, the total number of mobiles sold by Redmi = $B\%$ of 1000 = $10B$

Now,

$$(10C+50) + 10B = 40\% \text{ of } 1000$$

$$(10C+50) + 10B = 400$$

$$10C + 10B = 350$$

$$C + B = 35 \dots\dots\dots(I)$$

Also,

$$(10C+50) - 10B = 200$$

$$10C - 10B = 150$$

@QuizByMickey

$$C - B = 15 \dots\dots\dots(II)$$

Buy solving equation (I) and (II), we get,

$$2B = 20$$

$$B = 10$$

Now, put the value of B in equation (I), we get,

$$C + B = 35$$

$$C + 10 = 35$$

$$C = 25$$

Therefore,

$$\text{The total number of mobiles sold by Oppo} = (10C+50) = (10 \times 25 + 50) = (250+50) = 300$$

$$\text{And, the total number of mobiles sold by Redmi} = 10B = (10 \times 10) = 100$$

Vivo

$$\text{The total number of mobiles manufactured} = 250$$

$$\text{The total number of mobiles sold} = 200$$

$$\text{The total number of mobiles unsold} = (250-200) = 50$$

$$\text{The number of 4G mobiles unsold} = \{(50+10)/2\} = (60/2) = 30$$

$$\text{The number of 5G mobiles unsold} = \{(50-10)/2\} = (40/2) = 20$$

$$\text{So, the number of 4G mobiles sold} = \{20/1\} \times 6 = 120$$

$$\text{Therefore, the number of 5G mobiles sold} = (200-120) = 80$$

$$\text{Now, the total number of 4G mobiles manufactured} = (120+30) = 150$$

$$\text{And, the total number of 5G mobiles manufactured} = (80+20) = 100$$

Oppo

$$\text{The total number of mobiles manufactured} = 460$$

$$\text{The total number of mobiles sold} = 300$$

$$\text{The total number of mobiles unsold} = (460-300) = 160$$

$$\text{The number of 4G mobiles unsold} = \{(160+40)/2\} = (200/2) = 100$$

$$\text{The number of 5G mobiles unsold} = \{(160-40)/2\} = (120/2) = 60$$

$$\text{So, the number of 4G mobiles sold} = \{60/3\} \times 8 = 160$$

$$\text{Therefore, the number of 5G mobiles sold} = (300-160) = 140$$

$$\text{Now, the total number of 4G mobiles manufactured} = (160+100) = 260$$

$$\text{And, the total number of 5G mobiles manufactured} = (140+60) = 200$$

@QuizByMickey

Samsung

The total number of mobiles manufactured = 475

The total number of mobiles sold = 400

The total number of mobiles unsold = $(475-400) = 75$

The number of 4G mobiles unsold = $\{(75+25)/2\} = (100/2) = 50$

The number of 5G mobiles unsold = $\{(75-25)/2\} = (50/2) = 25$

So, the number of 4G mobiles sold = $\{25/1\} \times 12 = 300$

Therefore, the number of 5G mobiles sold = $(400-300) = 100$

Now, the total number of 4G mobiles manufactured = $(300+50) = 350$

And, the total number of 5G mobiles manufactured = $(100+25) = 125$

Redmi

The total number of mobiles manufactured = 155

The total number of mobiles sold = 100

The total number of mobiles unsold = $(155-100) = 55$

The number of 4G mobiles unsold = $\{(55+35)/2\} = (90/2) = 45$

The number of 5G mobiles unsold = $\{(55-35)/2\} = (20/2) = 10$

So, the number of 4G mobiles sold = $\{10/2\} \times 15 = 75$

Therefore, the number of 5G mobiles sold = $(100-75) = 25$

Now, the total number of 4G mobiles manufactured = $(75+45) = 120$

And, the total number of 5G mobiles manufactured = $(25+10) = 35$

4G Mobiles

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	150	120	30
Oppo	260	160	100
Samsung	350	300	50
Redmi	120	75	45

5G Mobiles

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	100	80	20
Oppo	200	140	60
Samsung	125	100	25
Redmi	35	25	10

Answer: B

Oppo

The total number of 4G mobiles sold (males + females) = 160

The number of 4G mobiles sold to males = $\frac{9}{16}$ of 160 = 90

So, the number of 4G mobiles sold to females = $\frac{7}{16}$ of 160 = 70

Now, the number of 5G mobiles sold to females = 80% of 70 = 56

The total number of 5G mobiles sold (males + females) = 140

So, the number of 5G mobiles sold to males = $(140 - 56) = 84$

Therefore,

P = the number of 5G mobiles sold to females = 56

Q = the number of 5G mobiles sold to males = 84

So, $P < Q$.

Hence, the correct answer is option B.

13. Questions

From the pie chart given above, we have,

The total value of the pie chart = 100%

The total number of mobiles sold by Vivo = $(A - 12)\%$ of $(30A + 40) = \{(30A^2 + 40A - 360A - 480)/100\} = \{(30A^2 - 320A - 480)/100\}$

Now,

$$\{(30A^2 - 320A - 480)/100\} = 200$$

$$30A^2 - 320A - 480 = 20000$$

$$30A^2 - 320A - 20480 = 0$$

$$3A^2 - 32A - 2048 = 0$$

$$3A^2 - 96A + 64A - 2048 = 0$$

$$3A(A-32) + 64(A-32) = 0$$

$$(3A+64)(A-32) = 0$$

$$A = -21.33 \text{ or } +32$$

So, $A = 32$ (positive integer)

Therefore, the total number of mobiles sold $(4G + 5G)$ by all four companies together $= (30A+40) = (30 \times 32 + 40) = (960 + 40) = 1000$

The total number of mobiles sold by Vivo $= 200$

The total number of mobiles sold by Samsung $= (A+8)\%$ of 1000 $= (32+8)\%$ of 1000 $= 40\%$ of 1000 $= 400$

Now, the percentage of the total number of mobiles sold by Vivo and Samsung together $=$

$$\{(A-12)+(A+8)\}\% = \{(32-12)+(32+8)\}\% = (20+40)\% = 60\%$$

So, the percentage of the total number of mobiles sold by Oppo and Redmi together $= (100-60)\% = 40\%$

The total number of mobiles sold by Oppo $= (C+5)\%$ of 1000 $= (10C+50)$

And, the total number of mobiles sold by Redmi $= B\%$ of 1000 $= 10B$

Now,

$$(10C+50) + 10B = 40\% \text{ of } 1000$$

$$(10C+50) + 10B = 400$$

$$10C + 10B = 350$$

$$C + B = 35 \dots\dots\dots(I)$$

Also,

$$(10C+50) - 10B = 200$$

$$10C - 10B = 150$$

$$C - B = 15 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$2B = 20$$

$$B = 10$$

Now, put the value of B in equation (I), we get,

$$C + B = 35$$

$$C + 10 = 35$$

$$C = 25$$

@QuizByMickey

Therefore,

The total number of mobiles sold by Oppo = $(10C+50) = (10 \times 25 + 50) = (250 + 50) = 300$

And, the total number of mobiles sold by Redmi = $10B = (10 \times 10) = 100$

Vivo

The total number of mobiles manufactured = 250

The total number of mobiles sold = 200

The total number of mobiles unsold = $(250 - 200) = 50$

The number of 4G mobiles unsold = $\{(50+10)/2\} = (60/2) = 30$

The number of 5G mobiles unsold = $\{(50-10)/2\} = (40/2) = 20$

So, the number of 4G mobiles sold = $\{20/1\} \times 6 = 120$

Therefore, the number of 5G mobiles sold = $(200 - 120) = 80$

Now, the total number of 4G mobiles manufactured = $(120 + 30) = 150$

And, the total number of 5G mobiles manufactured = $(80 + 20) = 100$

Oppo

The total number of mobiles manufactured = 460

The total number of mobiles sold = 300

The total number of mobiles unsold = $(460 - 300) = 160$

The number of 4G mobiles unsold = $\{(160+40)/2\} = (200/2) = 100$

The number of 5G mobiles unsold = $\{(160-40)/2\} = (120/2) = 60$

So, the number of 4G mobiles sold = $\{60/3\} \times 8 = 160$

Therefore, the number of 5G mobiles sold = $(300 - 160) = 140$

Now, the total number of 4G mobiles manufactured = $(160 + 100) = 260$

And, the total number of 5G mobiles manufactured = $(140 + 60) = 200$

Samsung

The total number of mobiles manufactured = 475

The total number of mobiles sold = 400

The total number of mobiles unsold = $(475 - 400) = 75$

The number of 4G mobiles unsold = $\{(75+25)/2\} = (100/2) = 50$

The number of 5G mobiles unsold = $\{(75-25)/2\} = (50/2) = 25$

So, the number of 4G mobiles sold = $\{25/1\} \times 12 = 300$

Therefore, the number of 5G mobiles sold = $(400 - 300) = 100$

@QuizByMickey

Now, the total number of 4G mobiles manufactured = $(300+50) = 350$

And, the total number of 5G mobiles manufactured = $(100+25) = 125$

Redmi

The total number of mobiles manufactured = 155

The total number of mobiles sold = 100

The total number of mobiles unsold = $(155-100) = 55$

The number of 4G mobiles unsold = $\{(55+35)/2\} = (90/2) = 45$

The number of 5G mobiles unsold = $\{(55-35)/2\} = (20/2) = 10$

So, the number of 4G mobiles sold = $\{10/2\} \times 15 = 75$

Therefore, the number of 5G mobiles sold = $(100-75) = 25$

Now, the total number of 4G mobiles manufactured = $(75+45) = 120$

And, the total number of 5G mobiles manufactured = $(25+10) = 35$

4G Mobiles

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	150	120	30
Oppo	260	160	100
Samsung	350	300	50
Redmi	120	75	45

5G Mobiles

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	100	80	20
Oppo	200	140	60
Samsung	125	100	25
Redmi	35	25	10

Answer: C

Redmi (4G mobiles)

The total number of mobiles manufactured = 120

The total number of mobiles sold = 75

The number of mobiles unsold = 45

The overall S.P. of all sold mobiles = Rs. 900000

So, the S.P. of each mobile = $(900000/75) = \text{Rs. } 12000$

And, the M.P. of each mobile = $(12000+2000) = \text{Rs. } 14000$

Therefore, the C.P. of each mobile = $\{(14000/175) \times 100\} = \text{Rs. } 8000$

Now,

$$700D - 400 = 8000$$

$$700D = 8400$$

$$D = 12$$

Therefore, the overall cost price of all 4G mobiles unsold by Redmi = $\{(700D-400) \times 45\} = \{(700 \times 12 - 400) \times 45\} = \{(8400 - 400) \times 45\} = (8000 \times 45) = \text{Rs. } 360000$.

Hence, the correct answer is option C.

14. Questions

From the pie chart given above, we have,

The total value of the pie chart = 100%

The total number of mobiles sold by Vivo = $(A-12)\%$ of $(30A+40) = \{(30A^2+40A-360A-480)/100\} = \{(30A^2-320A-480)/100\}$

Now,

$$\{(30A^2-320A-480)/100\} = 200$$

$$30A^2 - 320A - 480 = 20000$$

$$30A^2 - 320A - 20480 = 0$$

$$3A^2 - 32A - 2048 = 0$$

$$3A^2 - 96A + 64A - 2048 = 0$$

$$3A(A-32) + 64(A-32) = 0$$

$$(3A+64)(A-32) = 0$$

$$A = -21.33 \text{ or } +32$$

So, $A = 32$ (positive integer)

Therefore, the total number of mobiles sold (4G + 5G) by all four companies together = $(30A+40) = (30 \times 32 + 40) = (960 + 40) = 1000$

@QuizByMickey

The total number of mobiles sold by Vivo = 200

The total number of mobiles sold by Samsung = $(A+8)\%$ of 1000 = $(32+8)\%$ of 1000 = 40% of 1000 = 400

Now, the percentage of the total number of mobiles sold by Vivo and Samsung together =
 $\{(A-12)+(A+8)\}\% = \{(32-12)+(32+8)\}\% = (20+40)\% = 60\%$

So, the percentage of the total number of mobiles sold by Oppo and Redmi together = $(100-60)\% = 40\%$

The total number of mobiles sold by Oppo = $(C+5)\%$ of 1000 = $(10C+50)$

And, the total number of mobiles sold by Redmi = $B\%$ of 1000 = $10B$

Now,

$$(10C+50) + 10B = 40\% \text{ of } 1000$$

$$(10C+50) + 10B = 400$$

$$10C + 10B = 350$$

$$C + B = 35 \dots \dots \dots (I)$$

Also,

$$(10C+50) - 10B = 200$$

$$10C - 10B = 150$$

$$C - B = 15 \dots \dots \dots (II)$$

By solving equation (I) and (II), we get,

$$2B = 20$$

$$B = 10$$

Now, put the value of B in equation (I), we get,

$$C + B = 35$$

$$C + 10 = 35$$

$$C = 25$$

Therefore,

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The total number of mobiles sold by Oppo = $(10C+50) = (10 \times 25 + 50) = (250+50) = 300$

And, the total number of mobiles sold by Redmi = $10B = (10 \times 10) = 100$

Vivo

The total number of mobiles manufactured = 250

The total number of mobiles sold = 200

The total number of mobiles unsold = $(250-200) = 50$

The number of 4G mobiles unsold = $\{(50+10)/2\} = (60/2) = 30$

The number of 5G mobiles unsold = $\{(50-10)/2\} = (40/2) = 20$

So, the number of 4G mobiles sold = $\{20/1\} \times 6 = 120$

Therefore, the number of 5G mobiles sold = $(200-120) = 80$

Now, the total number of 4G mobiles manufactured = $(120+30) = 150$

And, the total number of 5G mobiles manufactured = $(80+20) = 100$

Oppo

The total number of mobiles manufactured = 460

The total number of mobiles sold = 300

The total number of mobiles unsold = $(460-300) = 160$

The number of 4G mobiles unsold = $\{(160+40)/2\} = (200/2) = 100$

The number of 5G mobiles unsold = $\{(160-40)/2\} = (120/2) = 60$

So, the number of 4G mobiles sold = $\{60/3\} \times 8 = 160$

Therefore, the number of 5G mobiles sold = $(300-160) = 140$

Now, the total number of 4G mobiles manufactured = $(160+100) = 260$

And, the total number of 5G mobiles manufactured = $(140+60) = 200$

Samsung

The total number of mobiles manufactured = 475

The total number of mobiles sold = 400

The total number of mobiles unsold = $(475-400) = 75$

The number of 4G mobiles unsold = $\{(75+25)/2\} = (100/2) = 50$

The number of 5G mobiles unsold = $\{(75-25)/2\} = (50/2) = 25$

So, the number of 4G mobiles sold = $\{25/1\} \times 12 = 300$

Therefore, the number of 5G mobiles sold = $(400-300) = 100$

Now, the total number of 4G mobiles manufactured = $(300+50) = 350$

And, the total number of 5G mobiles manufactured = $(100+25) = 125$

Redmi

The total number of mobiles manufactured = 155

The total number of mobiles sold = 100

The total number of mobiles unsold = $(155-100) = 55$

The number of 4G mobiles unsold = $\{(55+35)/2\} = (90/2) = 45$

The number of 5G mobiles unsold = $\{(55-35)/2\} = (20/2) = 10$

So, the number of 4G mobiles sold = $\{10/2\} \times 15 = 75$

Therefore, the number of 5G mobiles sold = $(100-75) = 25$

Now, the total number of 4G mobiles manufactured = $(75+45) = 120$

And, the total number of 5G mobiles manufactured = $(25+10) = 35$

4G Mobiles

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	150	120	30
Oppo	260	160	100
Samsung	350	300	50
Redmi	120	75	45

5G Mobiles

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	100	80	20
Oppo	200	140	60
Samsung	125	100	25
Redmi	35	25	10

Answer: A

The total number of mobiles sold by Samsung = 400

Apple

The total number of mobiles sold = 60% of 400 = 240

The number of 4G mobiles sold = $3/4$ of 240 = 180

The number of 5G mobiles sold = $1/4$ of 240 = 60

Let the number of 4G and 5G mobiles unsold be m and $2m$ respectively.

The ratio of the number of 4G and 5G mobiles manufactured = 2:1.

Now,

$$(180+m)/(60+2m) = 2/1$$

$$180 + m = 120 + 4m$$

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$$3m = 60$$

$$m = 20$$

Therefore, the total number of mobiles manufactured (4G + 5G) by Apple = $\{(180+m)+(60+2m)\} = (240+3m) = (240+3 \times 20) = (240+60) = 300$.

Hence, the correct answer is option A.

15. Questions

From the pie chart given above, we have,

The total value of the pie chart = 100%

The total number of mobiles sold by Vivo = $(A-12)\%$ of $(30A+40) = \{(30A^2+40A-360A-480)/100\} = \{(30A^2-320A-480)/100\}$

Now,

$$\{(30A^2-320A-480)/100\} = 200$$

$$30A^2 - 320A - 480 = 20000$$

$$30A^2 - 320A - 20480 = 0$$

$$3A^2 - 32A - 2048 = 0$$

$$3A^2 - 96A + 64A - 2048 = 0$$

$$3A(A-32) + 64(A-32) = 0$$

$$(3A+64)(A-32) = 0$$

$$A = -21.33 \text{ or } +32$$

So, $A = 32$ (positive integer)

Therefore, the total number of mobiles sold (4G + 5G) by all four companies together = $(30A+40) = (30 \times 32 + 40) = (960 + 40) = 1000$

The total number of mobiles sold by Vivo = 200

The total number of mobiles sold by Samsung = $(A+8)\%$ of 1000 = $(32+8)\%$ of 1000 = 40% of 1000 = 400

Now, the percentage of the total number of mobiles sold by Vivo and Samsung together = $\{(A-12)+(A+8)\}\% = \{(32-12)+(32+8)\}\% = (20+40)\% = 60\%$

So, the percentage of the total number of mobiles sold by Oppo and Redmi together = $(100-60)\% = 40\%$

The total number of mobiles sold by Oppo = $(C+5)\%$ of 1000 = $(10C+50)$

And, the total number of mobiles sold by Redmi = $B\%$ of 1000 = $10B$

Now,

$$(10C+50) + 10B = 40\% \text{ of } 1000$$

$$(10C+50) + 10B = 400$$

$$10C + 10B = 350$$

$$C + B = 35 \dots\dots\dots(I)$$

Also,

$$(10C+50) - 10B = 200$$

$$10C - 10B = 150$$

$$C - B = 15 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$2B = 20$$

$$B = 10$$

Now, put the value of B in equation (I), we get,

$$C + B = 35$$

$$C + 10 = 35$$

$$C = 25$$

Therefore,

$$\text{The total number of mobiles sold by Oppo} = (10C+50) = (10 \times 25 + 50) = (250 + 50) = 300$$

$$\text{And, the total number of mobiles sold by Redmi} = 10B = (10 \times 10) = 100$$

Vivo

$$\text{The total number of mobiles manufactured} = 250$$

$$\text{The total number of mobiles sold} = 200$$

$$\text{The total number of mobiles unsold} = (250 - 200) = 50$$

$$\text{The number of 4G mobiles unsold} = \{(50 + 10)/2\} = (60/2) = 30$$

$$\text{The number of 5G mobiles unsold} = \{(50 - 10)/2\} = (40/2) = 20$$

$$\text{So, the number of 4G mobiles sold} = \{20/1\} \times 6 = 120$$

$$\text{Therefore, the number of 5G mobiles sold} = (200 - 120) = 80$$

$$\text{Now, the total number of 4G mobiles manufactured} = (120 + 30) = 150$$

$$\text{And, the total number of 5G mobiles manufactured} = (80 + 20) = 100$$

Oppo

$$\text{The total number of mobiles manufactured} = 460$$

$$\text{The total number of mobiles sold} = 300$$

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The total number of mobiles unsold = $(460-300) = 160$

The number of 4G mobiles unsold = $\{(160+40)/2\} = (200/2) = 100$

The number of 5G mobiles unsold = $\{(160-40)/2\} = (120/2) = 60$

So, the number of 4G mobiles sold = $\{60/3\} \times 8 = 160$

Therefore, the number of 5G mobiles sold = $(300-160) = 140$

Now, the total number of 4G mobiles manufactured = $(160+100) = 260$

And, the total number of 5G mobiles manufactured = $(140+60) = 200$

Samsung

The total number of mobiles manufactured = 475

The total number of mobiles sold = 400

The total number of mobiles unsold = $(475-400) = 75$

The number of 4G mobiles unsold = $\{(75+25)/2\} = (100/2) = 50$

The number of 5G mobiles unsold = $\{(75-25)/2\} = (50/2) = 25$

So, the number of 4G mobiles sold = $\{25/1\} \times 12 = 300$

Therefore, the number of 5G mobiles sold = $(400-300) = 100$

Now, the total number of 4G mobiles manufactured = $(300+50) = 350$

And, the total number of 5G mobiles manufactured = $(100+25) = 125$

Redmi

The total number of mobiles manufactured = 155

The total number of mobiles sold = 100

The total number of mobiles unsold = $(155-100) = 55$

The number of 4G mobiles unsold = $\{(55+35)/2\} = (90/2) = 45$

The number of 5G mobiles unsold = $\{(55-35)/2\} = (20/2) = 10$

So, the number of 4G mobiles sold = $\{10/2\} \times 15 = 75$

Therefore, the number of 5G mobiles sold = $(100-75) = 25$

Now, the total number of 4G mobiles manufactured = $(75+45) = 120$

And, the total number of 5G mobiles manufactured = $(25+10) = 35$

4G Mobiles

@QuizByMickey

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	150	120	30
Oppo	260	160	100
Samsung	350	300	50
Redmi	120	75	45

5G Mobiles

Company	The total number of mobiles manufactured	The number of mobiles sold	The number of mobiles unsold
Vivo	100	80	20
Oppo	200	140	60
Samsung	125	100	25
Redmi	35	25	10

Answer: D

Samsung

The number of 4G mobiles unsold = 50

So, the number of 3G mobiles unsold = $Y\%$ of 50 = $0.50Y$

The number of 3G mobiles sold = 80

The total number of mobiles manufactured (3G + 4G + 5G) = $\{350+125+(2Y+20)\} = (495+2Y)$

And, the total number of 4G and 5G mobiles manufactured = $(350+125) = 475$

So, the total number of 3G mobiles manufactured = $\{(495+2Y)-475\} = (2Y+20)$

Now,

$$80 + 0.50Y = 2Y + 20$$

$$1.50Y = 60$$

$$Y = 40$$

Therefore, the value = $(Y/2) = (40/2) = 20$.

Hence, the correct answer is option D.

16. Questions

The ratio of the number of runs scored in Test in 2020 and 2021 = 5:4. Let it be 5m and 4m respectively.

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The number of balls faced in Test in 2020 = 640

The number of balls faced in Test in 2021 = $\{(640/160) \times 100\} = 400$

Now,

$$\{(4m/400) \times 100\} - \{(5m/640) \times 100\} = 35$$

$$(4m/4) - (25m/32) = 35$$

$$(32m - 25m)/32 = 35$$

$$7m = 1120$$

$$m = 160$$

Also,

Let the number of runs scored in ODIs in 2022 and 2023 be $9n$ and $16n$ respectively.

The number of balls faced in ODIs in 2023 = 100

The number of balls faced in ODIs in 2022 = $(100 - 40) = 60$

Now,

$$\{(9n/60) \times 100\} + \{(16n/100) \times 100\} = 620$$

$$(45n/3) + 16n = 620$$

$$(45n + 48n)/3 = 620$$

$$93n = 1860$$

$$n = 20$$

2020

The total number of runs scored (Test + ODIs) = $(600 \times 2) = 1200$

The total number of balls faced (Test + ODIs) = $(370 \times 2) = 740$

The number of runs scored in Test = $5m = (5 \times 160) = 800$

The number of balls in Test = 640

The strike-rate in Test = $\{(800/640) \times 100\} = 125$

The number of runs scored in ODIs = $(1200 - 800) = 400$

The number of balls in ODIs = $(740 - 640) = 100$

The strike-rate in ODIs = $\{400/100\} \times 100 = 400$

2021

The total number of runs scored (Test + ODIs) = $(500 \times 2) = 1000$

The total number of balls faced (Test + ODIs) = $(320 \times 2) = 640$

The number of runs scored in Test = $4m = (4 \times 160) = 640$

@QuizByMickey

The number of balls in Test = 400

The strike-rate in Test = $\{(640/400) \times 100\} = 160$

The number of runs scored in ODIs = $(1000-640) = 360$

The number of balls in ODIs = $(640-400) = 240$

The strike-rate in ODIs = $\{600/400\} \times 100 = 150$

2022

The total number of runs scored (Test + ODIs) = $(340 \times 2) = 680$

The total number of balls faced (Test + ODIs) = $(280 \times 2) = 560$

The number of runs scored in ODIs = $9n = (9 \times 20) = 180$

The number of balls in ODIs = 60

The strike-rate in ODIs = $\{(180/60) \times 100\} = 300$

The number of runs scored in Test = $(680-180) = 500$

The number of balls in Test = $(560-60) = 500$

The strike-rate in Test = $\{500/500\} \times 100 = 100$

2023

The total number of runs scored (Test + ODIs) = $(400 \times 2) = 800$

The total number of balls faced (Test + ODIs) = $(250 \times 2) = 500$

The number of runs scored in ODIs = $16n = (16 \times 20) = 320$

The number of balls in ODIs = 100

The strike-rate in ODIs = $\{(320/100) \times 100\} = 320$

The number of runs scored in Test = $(800-320) = 480$

The number of balls in Test = $(500-100) = 400$

The strike-rate in Test = $\{480/400\} \times 100 = 120$

Test

Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	800	640	125
2021	640	400	160
2022	500	500	100
2023	480	400	120
Total	2420	1940	124.74 (Overall)

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ODI

Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	400	100	400
2021	360	240	150
2022	180	60	300
2023	320	100	320
Total	1260	500	252 (Overall)

Answer: D

Now, according to the question,

2020 (ODIs)

The total number of runs scored = 400

The total number of balls faced = 100

The number of dot balls faced = 4A

So, the remaining number of balls faced = (100-4A)

Now,

$$\{400/(100-4A)\} \times 100 = 100A$$

$$400/(100-4A) = 100A/100$$

$$10000A - 400A^2 = 40000$$

$$400A^2 - 10000A + 40000 = 0$$

$$A^2 - 25A + 100 = 0$$

$$A^2 - 20A - 5A + 100 = 0$$

$$A(A-20) - 5(A-20) = 0$$

$$(A-5)(A-20) = 0$$

$$A = +5 \text{ or } +20$$

So, $A \neq 20$

Therefore, $A = 5$ (single-digit natural number)

So, the number of dot balls faced = $4A = (4 \times 5) = 20$

The required percentage = $\{(20/100) \times 100\} \% = 20\%$.

Hence, the correct answer is option D.

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17. Questions

The ratio of the number of runs scored in Test in 2020 and 2021 = 5:4. Let it be 5m and 4m respectively.

The number of balls faced in Test in 2020 = 640

The number of balls faced in Test in 2021 = $\{(640/160) \times 100\} = 400$

Now,

$$\{(4m/400) \times 100\} - \{(5m/640) \times 100\} = 35$$

$$(4m/4) - (25m/32) = 35$$

$$(32m - 25m)/32 = 35$$

$$7m = 1120$$

$$m = 160$$

Also,

Let the number of runs scored in ODIs in 2022 and 2023 be 9n and 16n respectively.

The number of balls faced in ODIs in 2023 = 100

The number of balls faced in ODIs in 2022 = $(100 - 40) = 60$

Now,

$$\{(9n/60) \times 100\} + \{(16n/100) \times 100\} = 620$$

$$(45n/3) + 16n = 620$$

$$(45n + 48n)/3 = 620$$

$$93n = 1860$$

$$n = 20$$

2020

The total number of runs scored (Test + ODIs) = $(600 \times 2) = 1200$

The total number of balls faced (Test + ODIs) = $(370 \times 2) = 740$

The number of runs scored in Test = $5m = (5 \times 160) = 800$

The number of balls in Test = 640

The strike-rate in Test = $\{(800/640) \times 100\} = 125$

The number of runs scored in ODIs = $(1200 - 800) = 400$

The number of balls in ODIs = $(740 - 640) = 100$

The strike-rate in ODIs = $\{400/100\} \times 100 = 400$

2021

The total number of runs scored (Test + ODIs) = $(500 \times 2) = 1000$

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The total number of balls faced (Test + ODIs) = $(320 \times 2) = 640$

The number of runs scored in Test = $4m = (4 \times 160) = 640$

The number of balls in Test = 400

The strike-rate in Test = $\{(640/400) \times 100\} = 160$

The number of runs scored in ODIs = $(1000 - 640) = 360$

The number of balls in ODIs = $(640 - 400) = 240$

The strike-rate in ODIs = $\{600/400\} \times 100 = 150$

2022

The total number of runs scored (Test + ODIs) = $(340 \times 2) = 680$

The total number of balls faced (Test + ODIs) = $(280 \times 2) = 560$

The number of runs scored in ODIs = $9n = (9 \times 20) = 180$

The number of balls in ODIs = 60

The strike-rate in ODIs = $\{(180/60) \times 100\} = 300$

The number of runs scored in Test = $(680 - 180) = 500$

The number of balls in Test = $(560 - 60) = 500$

The strike-rate in Test = $\{500/500\} \times 100 = 100$

2023

The total number of runs scored (Test + ODIs) = $(400 \times 2) = 800$

The total number of balls faced (Test + ODIs) = $(250 \times 2) = 500$

The number of runs scored in ODIs = $16n = (16 \times 20) = 320$

The number of balls in ODIs = 100

The strike-rate in ODIs = $\{(320/100) \times 100\} = 320$

The number of runs scored in Test = $(800 - 320) = 480$

The number of balls in Test = $(500 - 100) = 400$

The strike-rate in Test = $\{480/400\} \times 100 = 120$

Test

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Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	800	640	125
2021	640	400	160
2022	500	500	100
2023	480	400	120
Total	2420	1940	124.74 (Overall)

ODI

Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	400	100	400
2021	360	240	150
2022	180	60	300
2023	320	100	320
Total	1260	500	252 (Overall)

Answer: C

2023 (Test)

The total number of runs scored = 480

The number of innings played = X

So, the actual average = $(480/X)$

But, after 4 more innings, the average = $\{(480+200)/(X+4)\} = \{680/(X+4)\}$

Now,

$$\{680/(X+4)\} - (480/X) = 4$$

$$\{680X - (480X + 1920)/(X^2 + 4X) = 4$$

$$200X - 1920 = 4X^2 + 16X$$

$$4X^2 - 184X + 1920 = 0$$

$$X^2 - 46X + 480 = 0$$

$$X^2 - 30X - 16X + 480 = 0$$

$$X(X-30) - 16(X-30) = 0$$

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$$(X-16)(X-30) = 0$$

$$X = +16 \text{ or } +30$$

So, $X = 16$ (perfect square)

The total number of innings played = $X = 16$

Therefore, the total number of matches played in Test in 2023 = $(16/2) = 8$.

Hence, the correct answer is option C.

18. Questions

The ratio of the number of runs scored in Test in 2020 and 2021 = 5:4. Let it be $5m$ and $4m$ respectively.

The number of balls faced in Test in 2020 = 640

The number of balls faced in Test in 2021 = $\{(640/160) \times 100\} = 400$

Now,

$$\{(4m/400) \times 100\} - \{(5m/640) \times 100\} = 35$$

$$(4m/4) - (25m/32) = 35$$

$$(32m - 25m)/32 = 35$$

$$7m = 1120$$

$$m = 160$$

Also,

Let the number of runs scored in ODIs in 2022 and 2023 be $9n$ and $16n$ respectively.

The number of balls faced in ODIs in 2023 = 100

The number of balls faced in ODIs in 2022 = $(100-40) = 60$

Now,

$$\{(9n/60) \times 100\} + \{(16n/100) \times 100\} = 620$$

$$(45n/3) + 16n = 620$$

$$(45n + 48n)/3 = 620$$

$$93n = 1860$$

$$n = 20$$

2020

The total number of runs scored (Test + ODIs) = $(600 \times 2) = 1200$

The total number of balls faced (Test + ODIs) = $(370 \times 2) = 740$

The number of runs scored in Test = $5m = (5 \times 160) = 800$

The number of balls in Test = 640

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The strike-rate in Test = $\{(800/640) \times 100\} = 125$

The number of runs scored in ODIs = $(1200-800) = 400$

The number of balls in ODIs = $(740-640) = 100$

The strike-rate in ODIs = $\{400/100\} \times 100 = 400$

2021

The total number of runs scored (Test + ODIs) = $(500 \times 2) = 1000$

The total number of balls faced (Test + ODIs) = $(320 \times 2) = 640$

The number of runs scored in Test = $4m = (4 \times 160) = 640$

The number of balls in Test = 400

The strike-rate in Test = $\{(640/400) \times 100\} = 160$

The number of runs scored in ODIs = $(1000-640) = 360$

The number of balls in ODIs = $(640-400) = 240$

The strike-rate in ODIs = $\{360/240\} \times 100 = 150$

2022

The total number of runs scored (Test + ODIs) = $(340 \times 2) = 680$

The total number of balls faced (Test + ODIs) = $(280 \times 2) = 560$

The number of runs scored in ODIs = $9n = (9 \times 20) = 180$

The number of balls in ODIs = 60

The strike-rate in ODIs = $\{(180/60) \times 100\} = 300$

The number of runs scored in Test = $(680-180) = 500$

The number of balls in Test = $(560-60) = 500$

The strike-rate in Test = $\{500/500\} \times 100 = 100$

2023

The total number of runs scored (Test + ODIs) = $(400 \times 2) = 800$

The total number of balls faced (Test + ODIs) = $(250 \times 2) = 500$

The number of runs scored in ODIs = $16n = (16 \times 20) = 320$

The number of balls in ODIs = 100

The strike-rate in ODIs = $\{(320/100) \times 100\} = 320$

The number of runs scored in Test = $(800-320) = 480$

The number of balls in Test = $(500-100) = 400$

The strike-rate in Test = $\{480/400\} \times 100 = 120$

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Test

Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	800	640	125
2021	640	400	160
2022	500	500	100
2023	480	400	120
Total	2420	1940	124.74 (Overall)

ODI

Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	400	100	400
2021	360	240	150
2022	180	60	300
2023	320	100	320
Total	1260	500	252 (Overall)

Answer: A

2021 (ODIs)

The total number of runs scored = 360

The number of runs scored from sixes = $(P \times 6) = 6P$

The number of runs scored from fours = $\{(P \times 5) \times 4\} = (5P \times 4) = 20P$

So, the total number of runs scored in boundaries (fours + sixes) = $(20P + 6P) = 26P$

The number of runs scored other than boundaries = $(360 - 26P)$

Now,

$$26P - (360 - 26P) = 16P$$

$$10P = 360 - 26P$$

$$36P = 360$$

$$P = 10$$

So, 10 is a composite number.

Therefore, ? = composite

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Hence, the correct answer is option A.

19. Questions

The ratio of the number of runs scored in Test in 2020 and 2021 = 5:4. Let it be 5m and 4m respectively.

The number of balls faced in Test in 2020 = 640

The number of balls faced in Test in 2021 = $\{(640/160) \times 100\} = 400$

Now,

$$\{(4m/400) \times 100\} - \{(5m/640) \times 100\} = 35$$

$$(4m/4) - (25m/32) = 35$$

$$(32m - 25m)/32 = 35$$

$$7m = 1120$$

$$m = 160$$

Also,

Let the number of runs scored in ODIs in 2022 and 2023 be 9n and 16n respectively.

The number of balls faced in ODIs in 2023 = 100

The number of balls faced in ODIs in 2022 = $(100 - 40) = 60$

Now,

$$\{(9n/60) \times 100\} + \{(16n/100) \times 100\} = 620$$

$$(45n/3) + 16n = 620$$

$$(45n + 48n)/3 = 620$$

$$93n = 1860$$

$$n = 20$$

2020

The total number of runs scored (Test + ODIs) = $(600 \times 2) = 1200$

The total number of balls faced (Test + ODIs) = $(370 \times 2) = 740$

The number of runs scored in Test = $5m = (5 \times 160) = 800$

The number of balls in Test = 640

The strike-rate in Test = $\{(800/640) \times 100\} = 125$

The number of runs scored in ODIs = $(1200 - 800) = 400$

The number of balls in ODIs = $(740 - 640) = 100$

The strike-rate in ODIs = $\{(400/100) \times 100\} = 400$

2021

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The total number of runs scored (Test + ODIs) = $(500 \times 2) = 1000$

The total number of balls faced (Test + ODIs) = $(320 \times 2) = 640$

The number of runs scored in Test = $4m = (4 \times 160) = 640$

The number of balls in Test = 400

The strike-rate in Test = $\{(640/400) \times 100\} = 160$

The number of runs scored in ODIs = $(1000 - 640) = 360$

The number of balls in ODIs = $(640 - 400) = 240$

The strike-rate in ODIs = $\{600/400\} \times 100 = 150$

2022

The total number of runs scored (Test + ODIs) = $(340 \times 2) = 680$

The total number of balls faced (Test + ODIs) = $(280 \times 2) = 560$

The number of runs scored in ODIs = $9n = (9 \times 20) = 180$

The number of balls in ODIs = 60

The strike-rate in ODIs = $\{(180/60) \times 100\} = 300$

The number of runs scored in Test = $(680 - 180) = 500$

The number of balls in Test = $(560 - 60) = 500$

The strike-rate in Test = $\{500/500\} \times 100 = 100$

2023

The total number of runs scored (Test + ODIs) = $(400 \times 2) = 800$

The total number of balls faced (Test + ODIs) = $(250 \times 2) = 500$

The number of runs scored in ODIs = $16n = (16 \times 20) = 320$

The number of balls in ODIs = 100

The strike-rate in ODIs = $\{(320/100) \times 100\} = 320$

The number of runs scored in Test = $(800 - 320) = 480$

The number of balls in Test = $(500 - 100) = 400$

The strike-rate in Test = $\{480/400\} \times 100 = 120$

Test

@QuizByMickey

Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	800	640	125
2021	640	400	160
2022	500	500	100
2023	480	400	120
Total	2420	1940	124.74 (Overall)

ODI

Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	400	100	400
2021	360	240	150
2022	180	60	300
2023	320	100	320
Total	1260	500	252 (Overall)

Answer: A

2022 (Test)

The number of runs scored by the given player = 500

So, the number of runs scored by another player = $\{500 + (Z\% \text{ of } 500)\} = (500 + 5Z)$

The number of balls faced by another player = 500

Now,

$$\{(500 + 5Z)/500\} \times 100 = (3Z + 20)$$

$$(500 + 5Z)/500 = (3Z + 20)/100$$

$$50000 + 500Z = 1500Z + 10000$$

$$1000Z = 40000$$

$$Z = 40$$

Therefore, the number of runs scored by another player in Test in 2022 = $(500 + 5Z) = (500 + 5 \times 40) = (500 + 200) = 700$.

Hence, the correct answer is option A.

20. Questions

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The ratio of the number of runs scored in Test in 2020 and 2021 = 5:4. Let it be 5m and 4m respectively.

The number of balls faced in Test in 2020 = 640

The number of balls faced in Test in 2021 = $\{(640/160) \times 100\} = 400$

Now,

$$\{(4m/400) \times 100\} - \{(5m/640) \times 100\} = 35$$

$$(4m/4) - (25m/32) = 35$$

$$(32m - 25m)/32 = 35$$

$$7m = 1120$$

$$m = 160$$

Also,

Let the number of runs scored in ODIs in 2022 and 2023 be 9n and 16n respectively.

The number of balls faced in ODIs in 2023 = 100

The number of balls faced in ODIs in 2022 = $(100 - 40) = 60$

Now,

$$\{(9n/60) \times 100\} + \{(16n/100) \times 100\} = 620$$

$$(45n/3) + 16n = 620$$

$$(45n + 48n)/3 = 620$$

$$93n = 1860$$

$$n = 20$$

2020

The total number of runs scored (Test + ODIs) = $(600 \times 2) = 1200$

The total number of balls faced (Test + ODIs) = $(370 \times 2) = 740$

The number of runs scored in Test = $5m = (5 \times 160) = 800$

The number of balls in Test = 640

The strike-rate in Test = $\{(800/640) \times 100\} = 125$

The number of runs scored in ODIs = $(1200 - 800) = 400$

The number of balls in ODIs = $(740 - 640) = 100$

The strike-rate in ODIs = $\{400/100\} \times 100 = 400$

2021

The total number of runs scored (Test + ODIs) = $(500 \times 2) = 1000$

The total number of balls faced (Test + ODIs) = $(320 \times 2) = 640$

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The number of runs scored in Test = $4m = (4 \times 160) = 640$

The number of balls in Test = 400

The strike-rate in Test = $\{(640/400) \times 100\} = 160$

The number of runs scored in ODIs = $(1000 - 640) = 360$

The number of balls in ODIs = $(640 - 400) = 240$

The strike-rate in ODIs = $\{600/400\} \times 100 = 150$

2022

The total number of runs scored (Test + ODIs) = $(340 \times 2) = 680$

The total number of balls faced (Test + ODIs) = $(280 \times 2) = 560$

The number of runs scored in ODIs = $9n = (9 \times 20) = 180$

The number of balls in ODIs = 60

The strike-rate in ODIs = $\{(180/60) \times 100\} = 300$

The number of runs scored in Test = $(680 - 180) = 500$

The number of balls in Test = $(560 - 60) = 500$

The strike-rate in Test = $\{500/500\} \times 100 = 100$

2023

The total number of runs scored (Test + ODIs) = $(400 \times 2) = 800$

The total number of balls faced (Test + ODIs) = $(250 \times 2) = 500$

The number of runs scored in ODIs = $16n = (16 \times 20) = 320$

The number of balls in ODIs = 100

The strike-rate in ODIs = $\{(320/100) \times 100\} = 320$

The number of runs scored in Test = $(800 - 320) = 480$

The number of balls in Test = $(500 - 100) = 400$

The strike-rate in Test = $\{480/400\} \times 100 = 120$

Test

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Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	800	640	125
2021	640	400	160
2022	500	500	100
2023	480	400	120
Total	2420	1940	124.74 (Overall)

ODI

Years	The total number of runs scored	The number of balls faced	The strike-rate
2020	400	100	400
2021	360	240	150
2022	180	60	300
2023	320	100	320
Total	1260	500	252 (Overall)

Answer: B

The total number of runs scored in 2023 = 800

2024

The total number of runs scored = 125% of 800 = 1000

Let the number of runs scored in Test be K.

So, the number of runs scored in ODIs = (1000-K)

The number of balls faced in ODIs = 125

So, the number of balls faced in Test = $\{(125/25) \times 100\} = 500$

Now,

$$[\{(1000-K)/125\} \times 100] - \{(K/500) \times 100\} = 50$$

$$\{(4000-4K)/5\} - (K/5) = 50$$

$$\{(4000-4K)-K\}/5 = 50$$

$$4000 - 5K = 250$$

$$5K = 3750$$

$$K = 750$$

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Now, the number of runs scored by the player in Test = $K = 750$

And, the number of runs scored by the player in ODIs = $(1000 - K) = (1000 - 750) = 250$

Therefore, the required ratio = $750:250 = 3:1$.

Hence, the correct answer is option B.

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